BICOL

Terrain & Signal Propagation

Albay and Sorsogon

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Photo: TheNuggeteer, Wikimedia Commons, CC BY-SA 4.0

Mission and analysis roadmap

MISSION

As the Signal Officer, establish and maintain reliable tactical communications in support of brigade operations by conducting terrain and signal propagation analysis, identifying communication dead spots, and recommending relay stations to ensure uninterrupted Command and Control (C2).

AO: Albay & Sorsogon • Bde HQ: Legazpi City

01 **Terrain analysis**
Mayon, the Pocdol Mountains and the seam between them

02 **Signal propagation**
Radio Mobile plots for all nine stations

03 **Line-of-sight**
12 planned links, every one checked

04 **Radio & cell coverage**
VHF plots plus Smart, Globe and DITO

05 **Dead spots**
Where the corridor loses VHF

06 **Relay station**
Relay Alpha at 51P XQ 01711 40097

07 **Antenna height & power**
Smallest mast and power that still work

BOTTOM LINE UP FRONT

One relay on Pocdol high ground joins Albay and Sorsogon into one brigade net

12 / 12

planned VHF links close in Radio Mobile

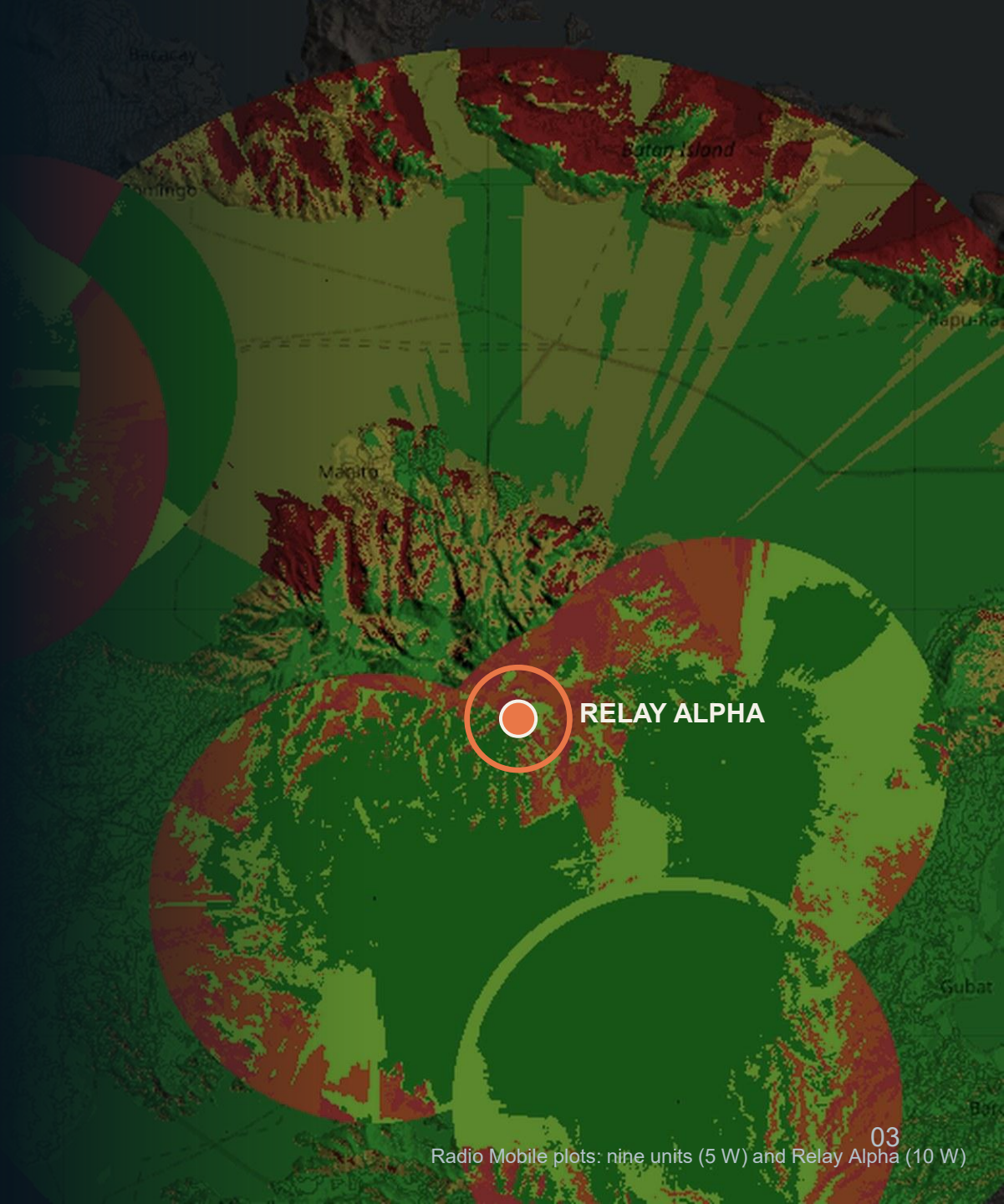
+35.8 dB

weakest fade margin: Daraga to Relay Alpha, 28.4 km

51P XQ 01711 40097

Relay Alpha: 931 m ground, 6 m mast, 10 W retransmission

Recommendation: emplace Relay Alpha with a security element, keep bases at 5 W, and confirm all 12 links by radio check.



Radio Mobile plots: nine units (5 W) and Relay Alpha (10 W)

STRATEGIC CONTEXT // 1 OF 3

From the archipelago to our AO

Republic of the Philippines — Bicol Region (Region V) highlighted.

109.0 M — Filipinos — 2020 census (PSA)

6.08 M — Bicol Region — 6 provinces

Typhoon belt — Bicol faces the Philippine Sea — first landfall for many storms



STRATEGIC CONTEXT // 2 OF 3

Bicol Region — our AO is its southern half

Albay and Sorsogon hold 2.20 M people — 36% of the region.

ALBAY

Legazpi City (Bde HQ) · Mayon Volcano

1,374,768	2,576 km²	18	720
people	land area	cities & mun.	barangays

SORSOGON

Sorsogon City · Matnog ferry to the Visayas

828,655	2,119 km²	15	541
people	land area	cities & mun.	barangays

Source: PSA 2020 Census of Population and Housing. Boundaries: NAMRIA/PSA via geoBoundaries.



STRATEGIC CONTEXT // 3 OF 3

33 cities and municipalities

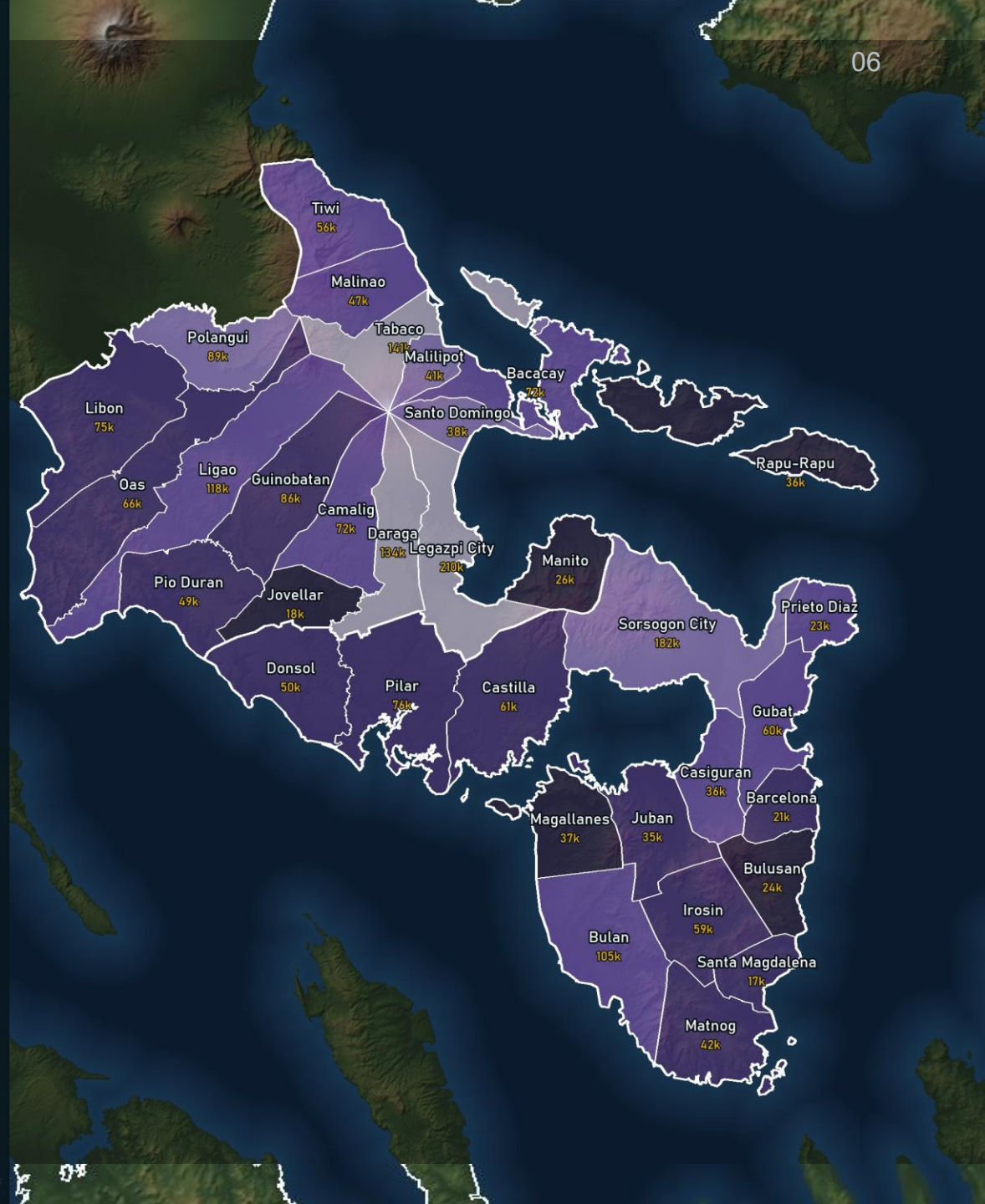
Population density by LGU (people per km², PSA 2020). Labels show each LGU's population.

LARGEST LGUs

Legazpi City	Albay	209,533
Sorsogon City	Sorsogon	182,237
Tabaco	Albay	140,961
Daraga	Albay	133,893
Ligao	Albay	118,096

PEOPLE PER KM²

< 250 | 250–400 | 400–600 | 600–1000 | > 1,000



Nine stations across two provinces



DISTANCE FROM BDE HQ

44 km

Farthest unit: Casiguran. Albay units sit 4–20 km from HQ; Sorsogon units 28–44 km, beyond the Pocdol Mountains.

LEGEND

- Brigade HQ (Legazpi)
- Albay units
- Sorsogon units

Station call signs and map key



Fox - DRG

Daraga



Wolf - GNB

Guinobatan



Eagle - LGP

Legazpi City TOC



Otter - CSG

Casiguran



Dolphin - SOR

Sorsogon City



Owl - CAM

Camalig



Stag - LIG

Ligao



Turtle - BAC

Bacon District



Bear - CST

Castilla



Falcon - RLY

Existing plotted relay

Full source place names remain in the map descriptions.

What the human and physical terrain means for comms



Mayon Volcano

Most active volcano in the country; 6 km Permanent Danger Zone. Ashfall and lahars threaten masts, generators and roads around Legazpi–Daraga–Camalig–Guinobatan.



Typhoon belt

Bicol faces the Philippine Sea. Expect grid power and cell-site outages after landfall: guy every mast, carry generator fuel, keep HF as alternate.



Urban clutter

Legazpi City (210k) and Sorsogon City (182k) are the densest LGUs. Handhelds lose 10–20 dB inside buildings — use vehicle or base sets for C2 there.



Commercial networks

Smart, Globe and DITO cover the towns and the coastal highway. Cellular is a contingency means only — encrypted apps, never classified traffic.



Language

Central Bikol across Albay; Sorsogon varieties in the south. Use local liaison for civil–military coordination and barangay reporting.



Lines of communication

The Maharlika Highway (AH26) is the main supply route, running to the Matnog ferry port for the Visayas. Plan retrans along it.



SECTION 01 // STUDENT TASK 1

01

Terrain Analysis

Where the ground helps, and where it blocks VHF line of sight.

Photo: Geabig2024, Wikimedia Commons, CC BY-SA 4.0

Regional terrain



Volcanic relief

Mayon’s steep slopes contrast with surrounding lower ground.

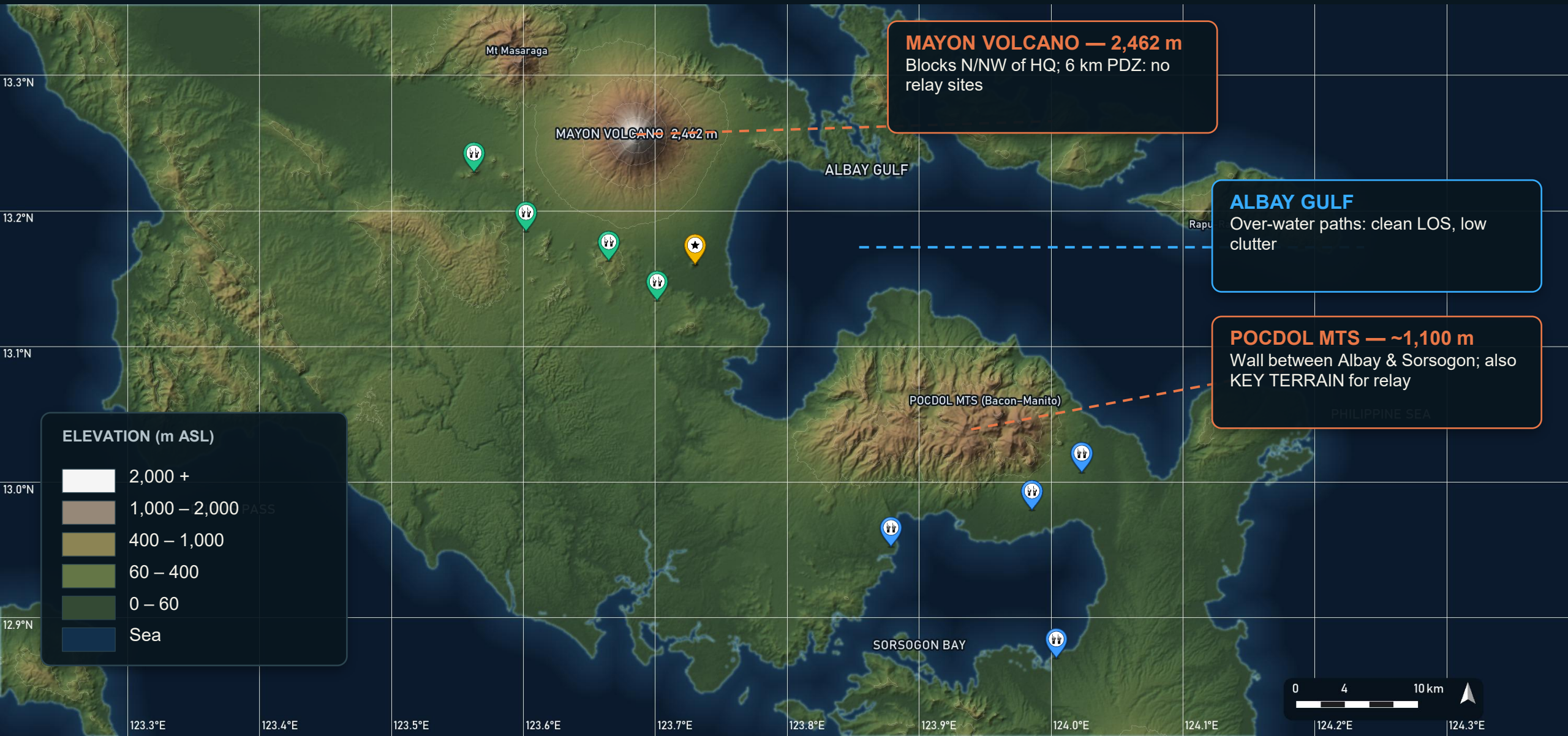
Hills, lowlands and coast

The project terrain view includes ridges, valleys and coastal areas.

Relief varies across the region. A radius alone cannot describe it.

Archival NASA image, December 2009. Geographic context, not a current hazard map.

Two massifs shape every radio path in the AO



MAYON VOLCANO — 2,462 m
Blocks N/NW of HQ; 6 km PDZ: no relay sites

ALBAY GULF
Over-water paths: clean LOS, low clutter

POCDOL MTS — ~1,100 m
Wall between Albay & Sorsogon; also KEY TERRAIN for relay

SECTION 02 // STUDENT TASK 2

02

Signal Propagation

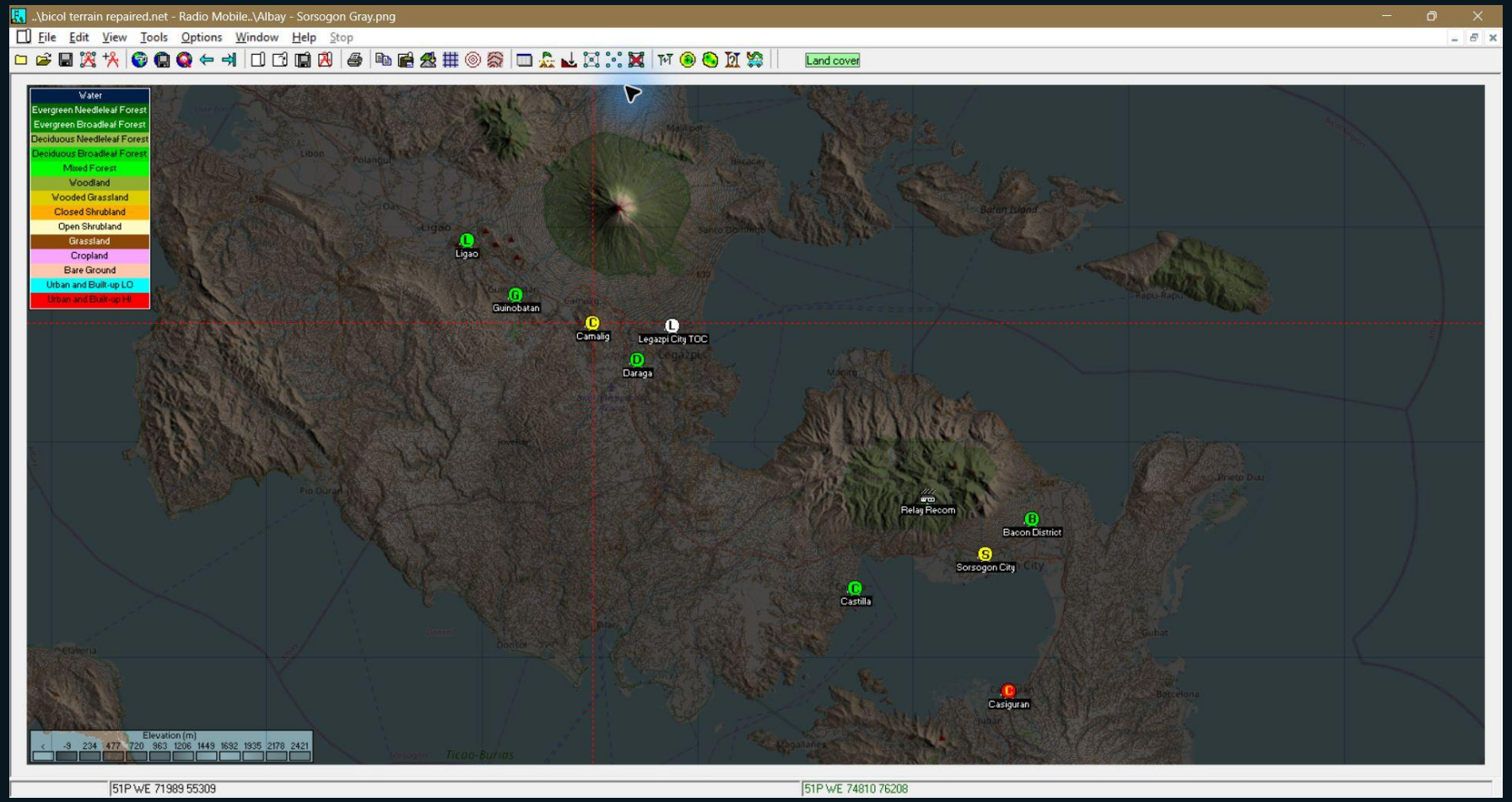
Radio Mobile point-to-area plots for all nine stations.



Photo: Pvt Adriel Paglinawan, Army Signal Regiment, Philippine Army (public domain)

ACTUAL RADIO MOBILE WORKSPACE

The working terrain project



Terrain relief, land-cover legend and station arrangement from your working project.

Radio Mobile parameters used for every plot

FREQUENCY

40–59 MHz

Harris handheld net (low VHF)

TX POWER

5 - 10 W

37–40 dBm (relay 10 W + 10 W)

RX THRESHOLD

−119 dBm

0.25 µV sensitivity

ANTENNA

4 dBi @ 6 m

HQ & relay 6 m, handhelds 2 m

LINE LOSS

0.5 dB

Each end

TERRAIN DATA

SRTM-3

90 m DEM

MODEL

Longley-Rice

ITM, k = 4/3

PLOT RADIUS

10 km

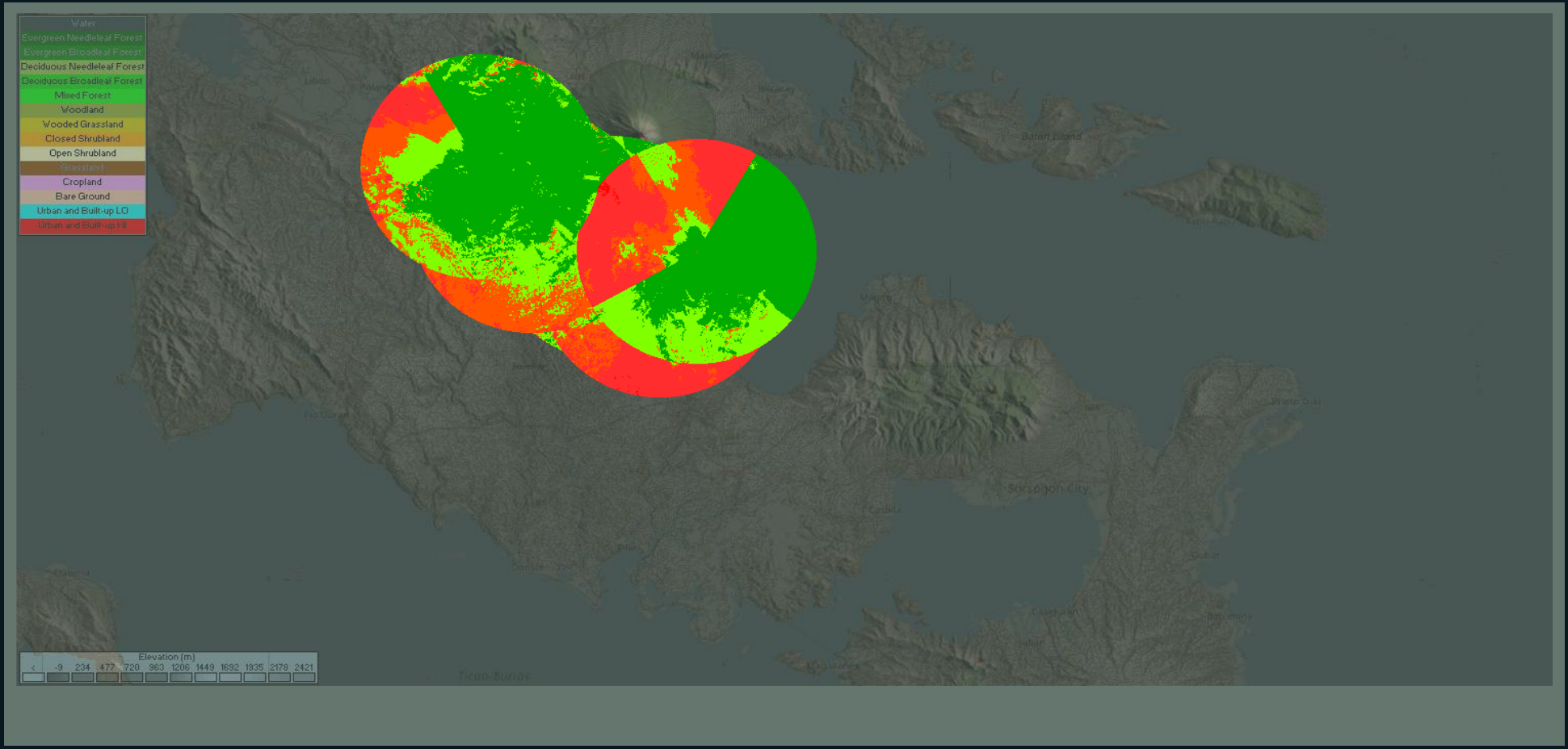
Per station

Link check
All 12 planned links computed in Radio Mobile Radio Link: 35.8 to 79.3 dB above the -119 dBm threshold (Section 03).

Radio settings in the saved captures

Input	LGP transmitter	Relay transmitter
Power	5 W	10 W
Antenna height	6 m	6 m
Antenna gain	4 dBi	4 dBi
Line loss	0.5 dB	0.5 dB
Frequency range	40–59 MHz	40–59 MHz

Albay radio overlays



Owl - CAM

Fox - DRG

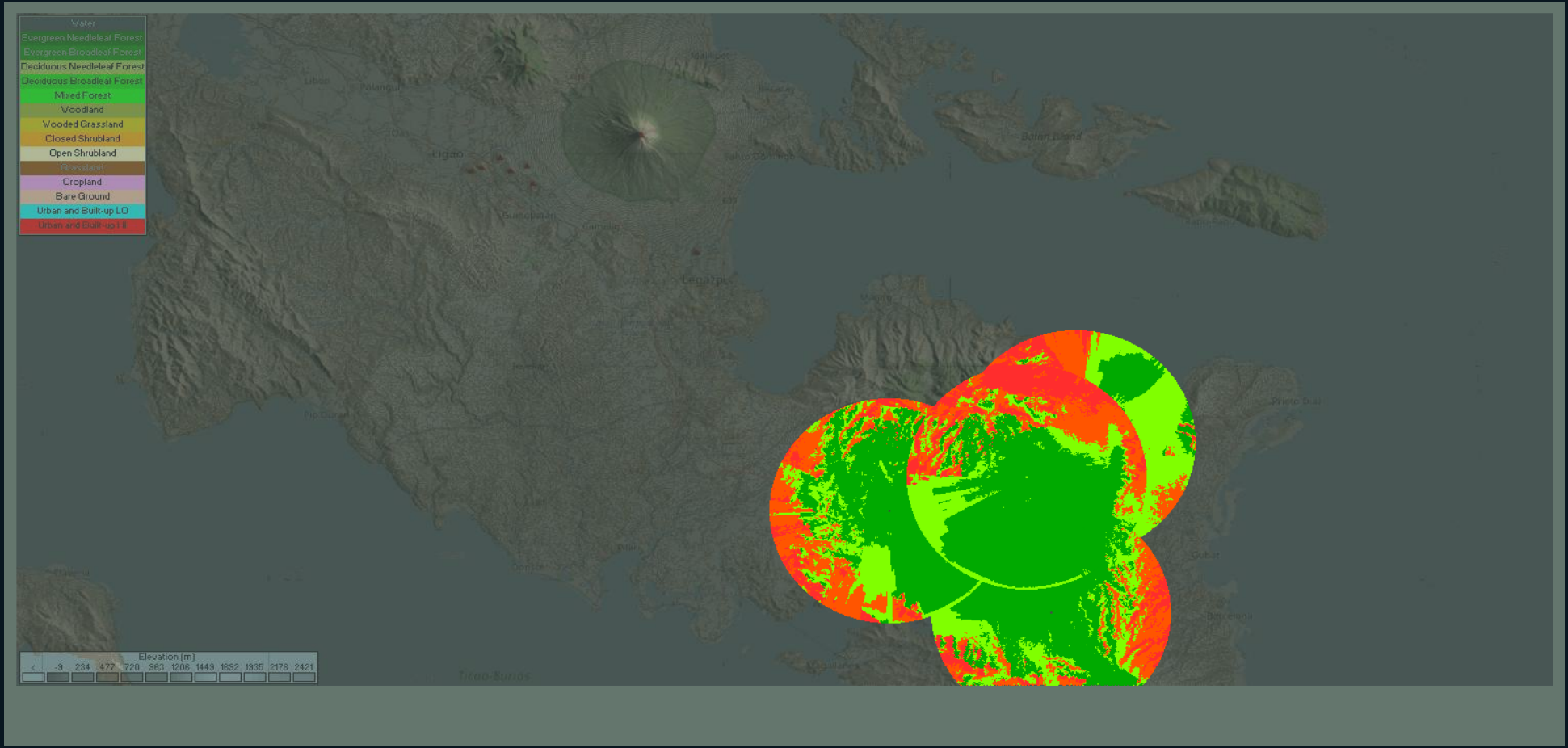
Wolf - GNB

Eagle - LGP

Stag - LIG

Original saved colour classes. Click to reveal the supplied overlays.

Sorsogon radio overlays



Turtle - BAC

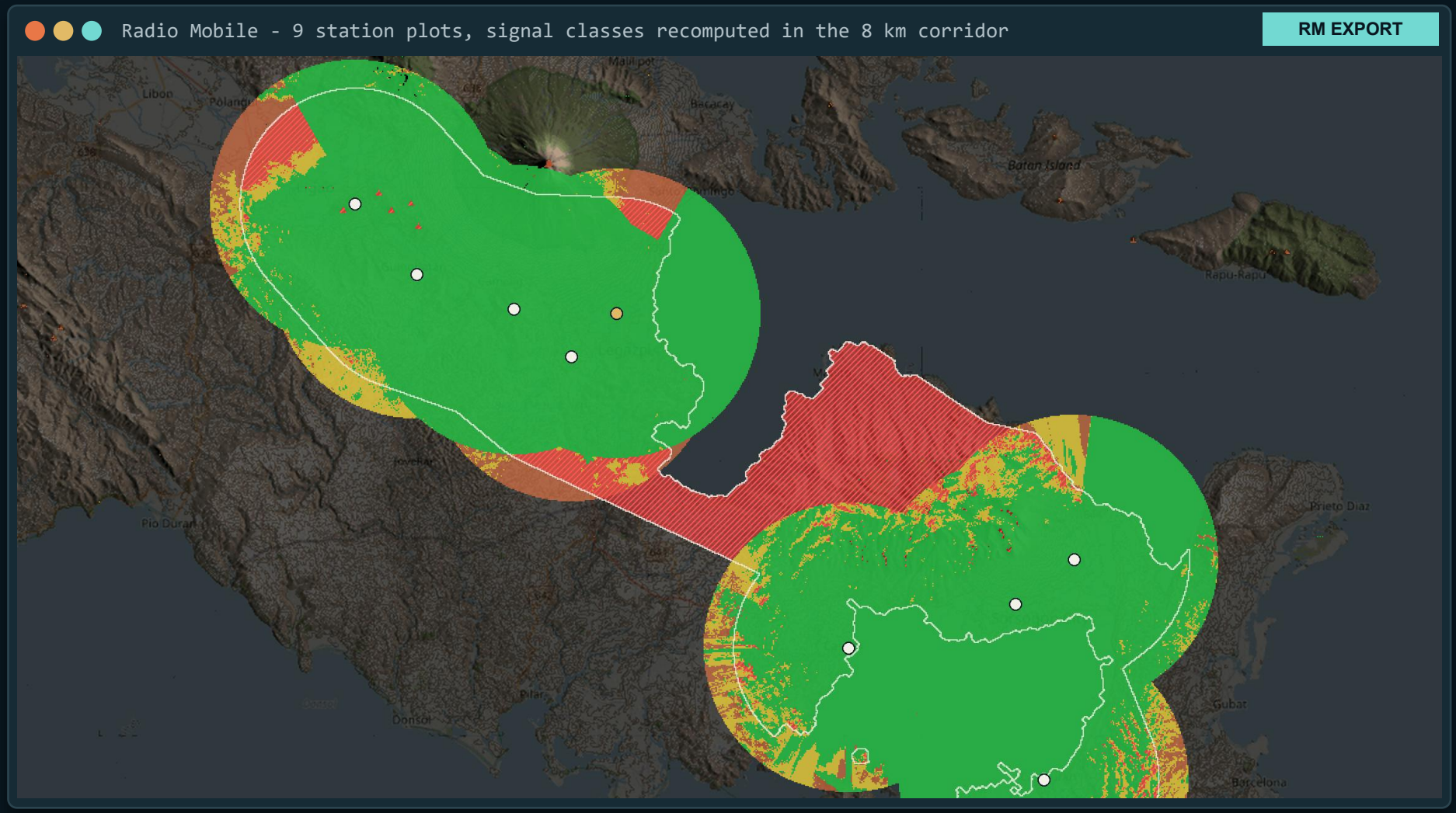
Otter - CSG

Bear - CST

Dolphin - SOR

Original saved colour classes. Click to reveal the supplied overlays.

Overall coverage — all nine stations (5 W)



CORRIDOR

81%

usable VHF (yellow or green)

75%

green class (20.5 and up)

The gap

Two islands of coverage. Between Daraga and Castilla 179 km² of corridor has no usable plot.

SECTION 03 // STUDENT TASK 3

03

Line-of-Sight

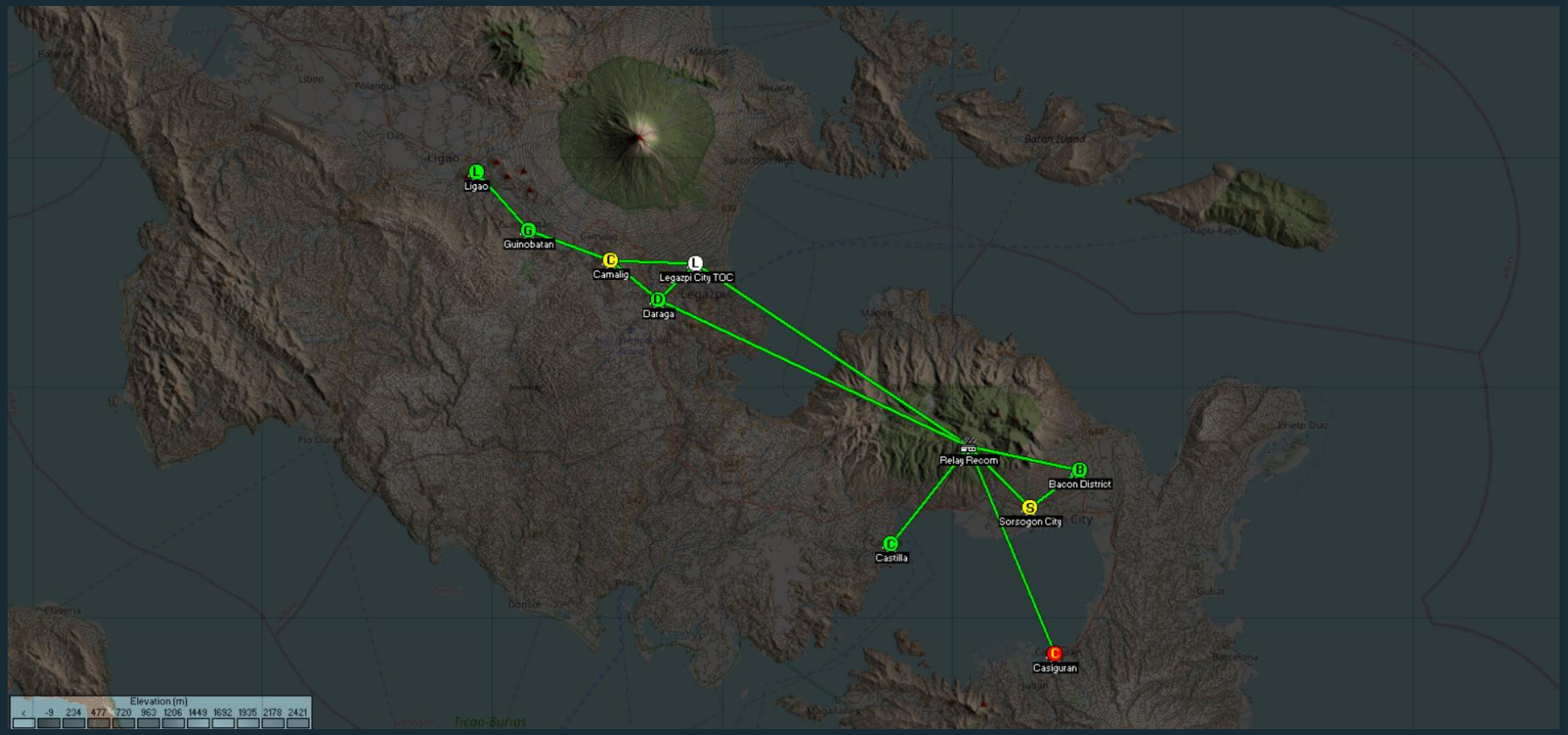
12 planned links, each checked in Radio Mobile Radio Link.

Photo: Irvin Parco Sto. Tomas, Wikimedia Commons, CC BY-SA 4.0

The network as built in Radio Mobile: 12 links

Radio Mobile - bicol terrain repaired.net: Albay, Repeater Connection and Sorsogon nets

SCREENSHOT



Links
Green lines are the 12 planned links in three Radio Mobile nets: Albay (5), Repeater Connection (2) and Sorsogon (5). Each one is checked on the next slides.

The supplied link map



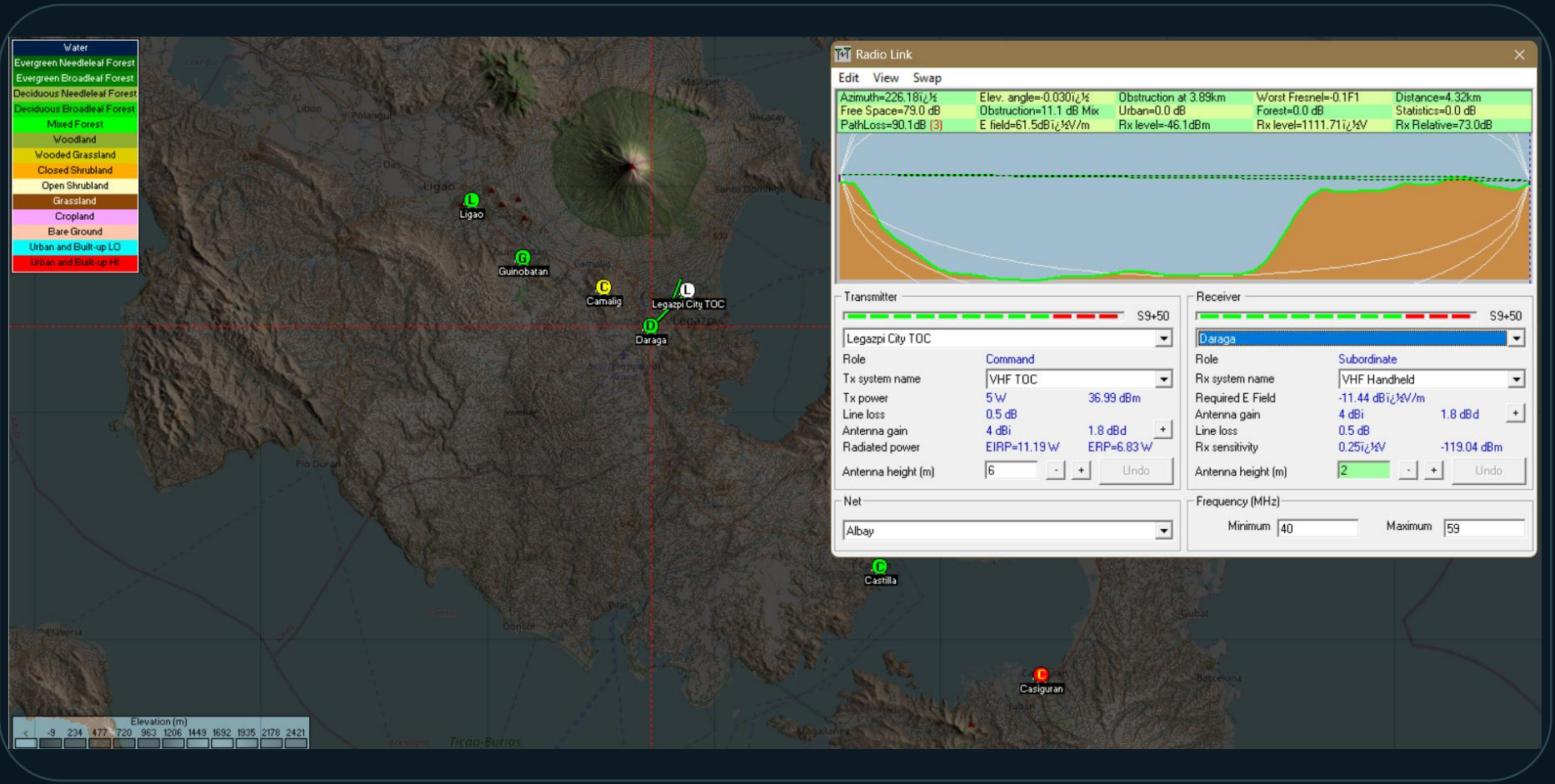
12

supplied beam paths

Includes the Guinobatan–Camalig and Sorsogon–Bacon source exports.

Albay net: five short links, all close by 53 dB or more

LGP > DRG | LGP > CAM | DRG > CAM | CAM > GNB | LIG > GNB



DISTANCE
4.32 km

RX LEVEL
-46.1 dBm

FADE MARGIN
+73.0 dB

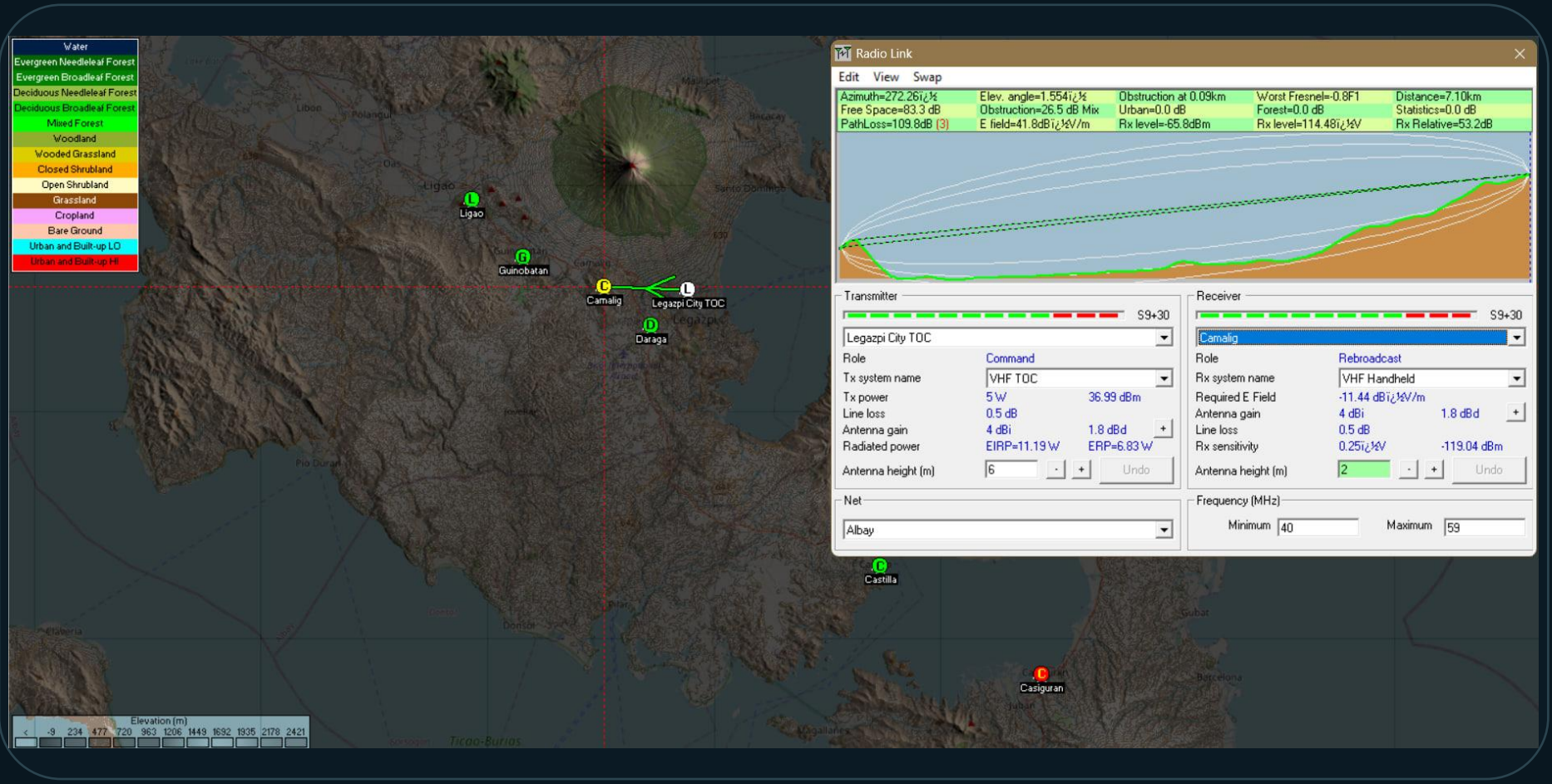
WORST FRESNEL
-0.1 F1

LINK GO
Terrain in path - closes by diffraction

Radio Mobile Radio Link | Tx Legazpi City TOC (Command) > Rx Daraga (Subordinate) | Albay net

Albay net: five short links, all close by 53 dB or more

LGP > DRG | **LGP > CAM** | DRG > CAM | CAM > GNB | LIG > GNB



DISTANCE
7.10 km

RX LEVEL
-65.8 dBm

FADE MARGIN
+53.2 dB

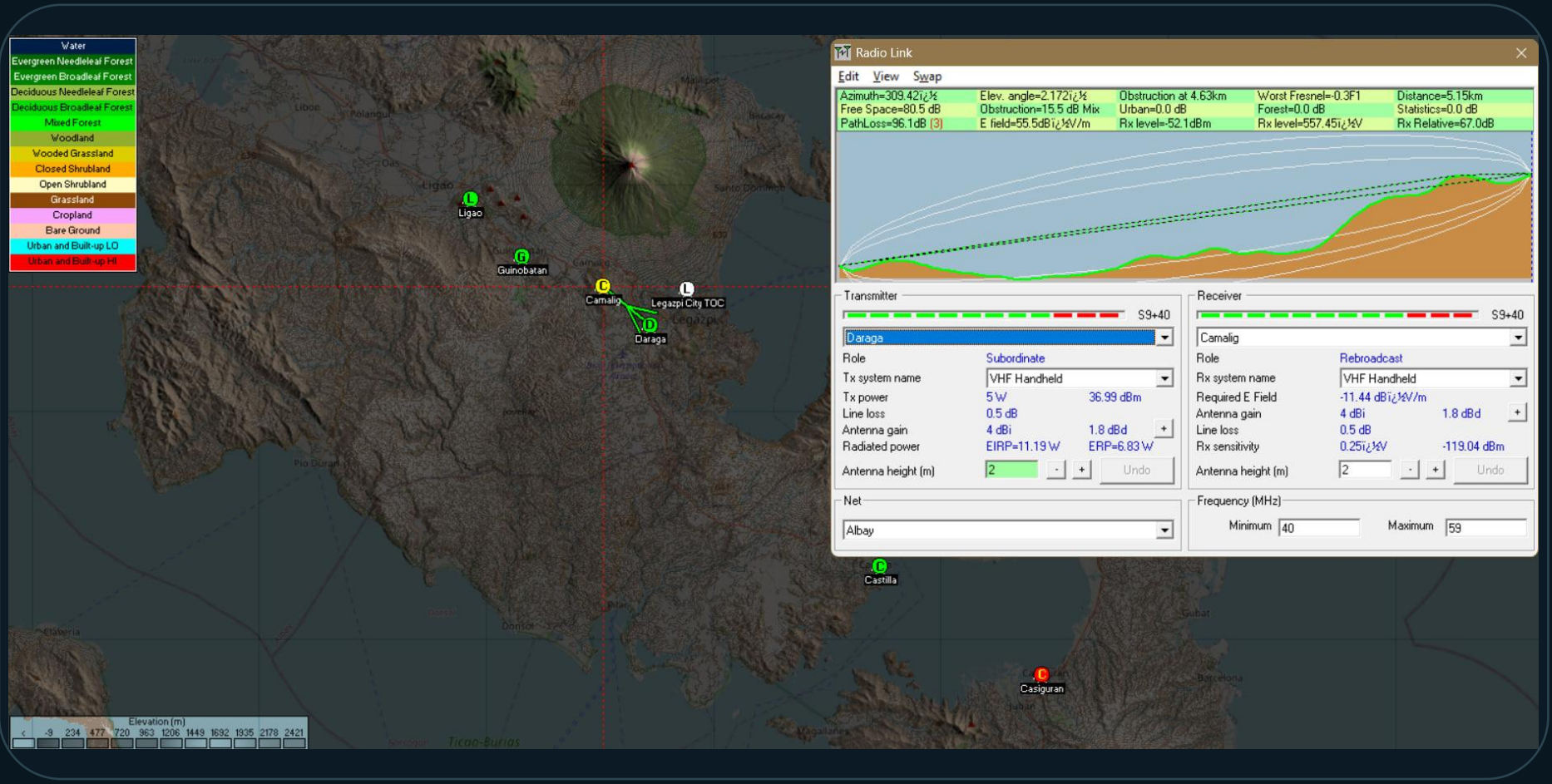
WORST FRESNEL
-0.8 F1

LINK GO
Terrain in path - closes by diffraction

Radio Mobile Radio Link | Tx Legazpi City TOC (Command) > Rx Camalig (Rebroadcast) | Albay net

Albay net: five short links, all close by 53 dB or more

LGP > DRG | LGP > CAM | **DRG > CAM** | CAM > GNB | LIG > GNB



DISTANCE
5.15 km

RX LEVEL
-52.1 dBm

FADE MARGIN
+67.0 dB

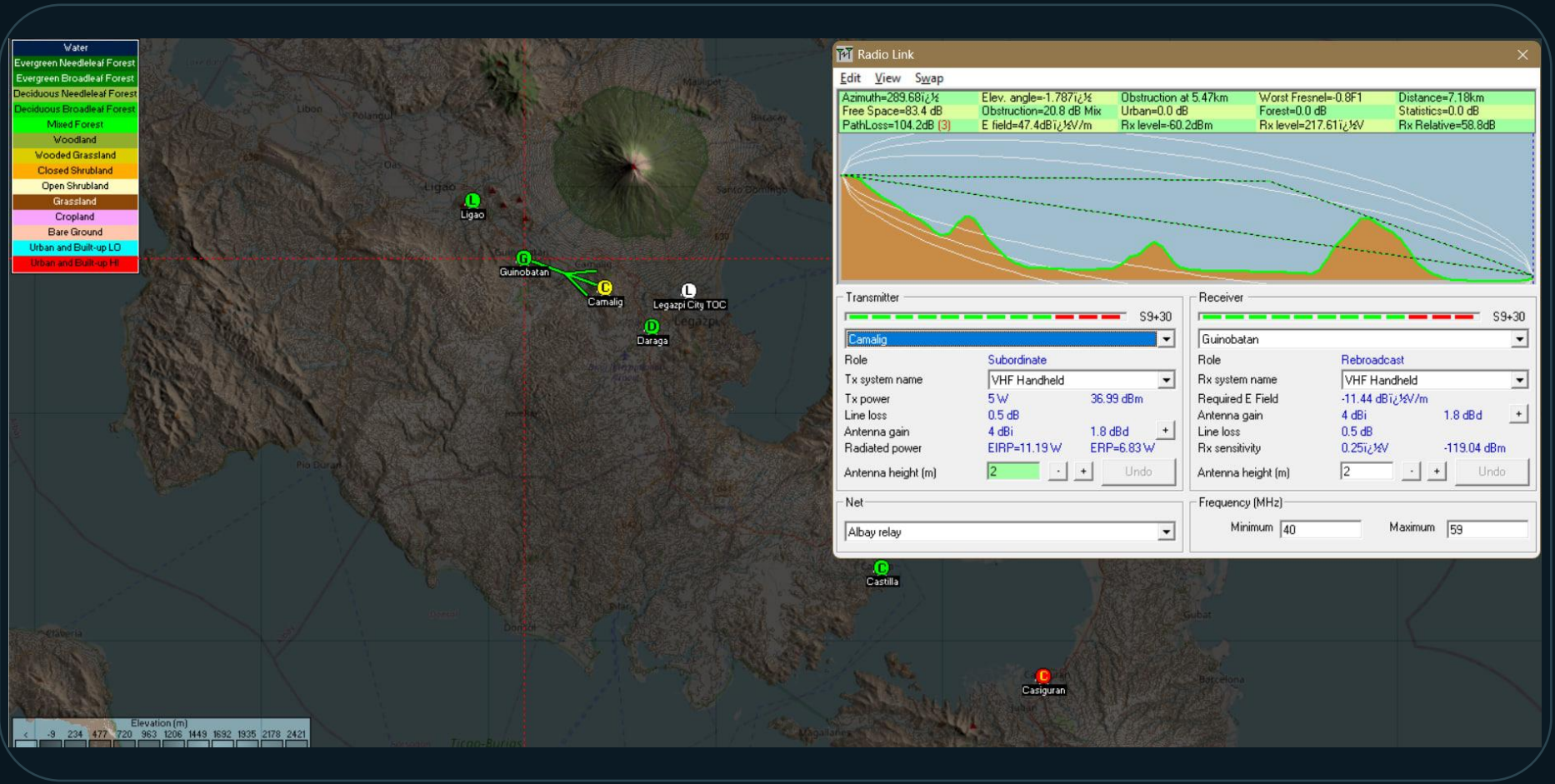
WORST FRESNEL
-0.3 F1

LINK GO
Terrain in path - closes by diffraction

Radio Mobile Radio Link | Tx Daraga (Subordinate) > Rx Camalig (Rebroadcast) | Albay net

Albay net: five short links, all close by 53 dB or more

LGP > DRG | LGP > CAM | DRG > CAM | **CAM > GNB** | LIG > GNB



DISTANCE
7.18 km

RX LEVEL
-60.2 dBm

FADE MARGIN
+58.8 dB

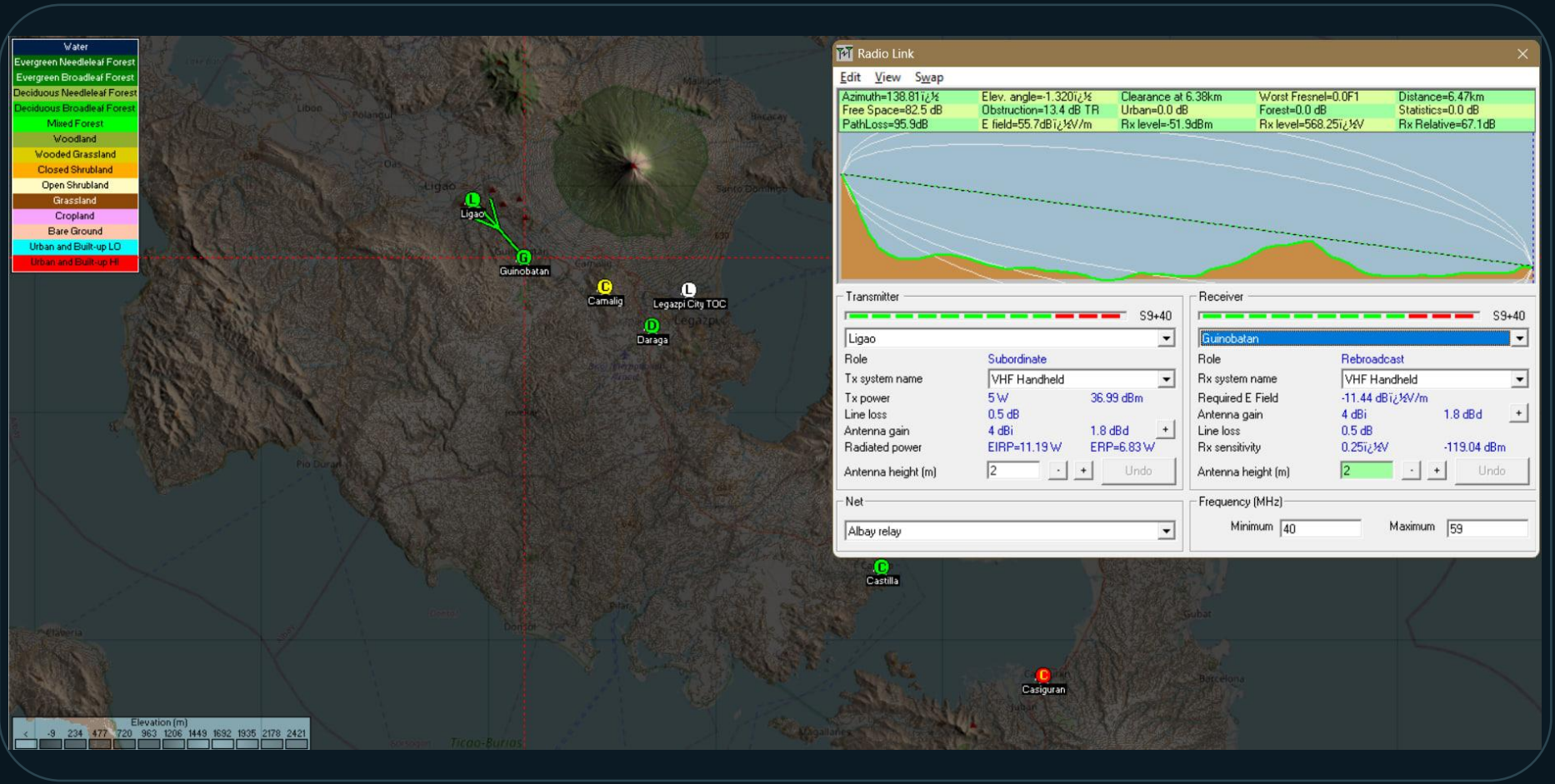
WORST FRESNEL
-0.8 F1

LINK GO
Terrain in path - closes by diffraction

Radio Mobile Radio Link | Tx Camalig (Subordinate) > Rx Guinobatan (Rebroadcast) | Albay net

Albay net: five short links, all close by 53 dB or more

LGP > DRG | LGP > CAM | DRG > CAM | CAM > GNB | **LIG > GNB**



DISTANCE
6.47 km

RX LEVEL
-51.9 dBm

FADE MARGIN
+67.1 dB

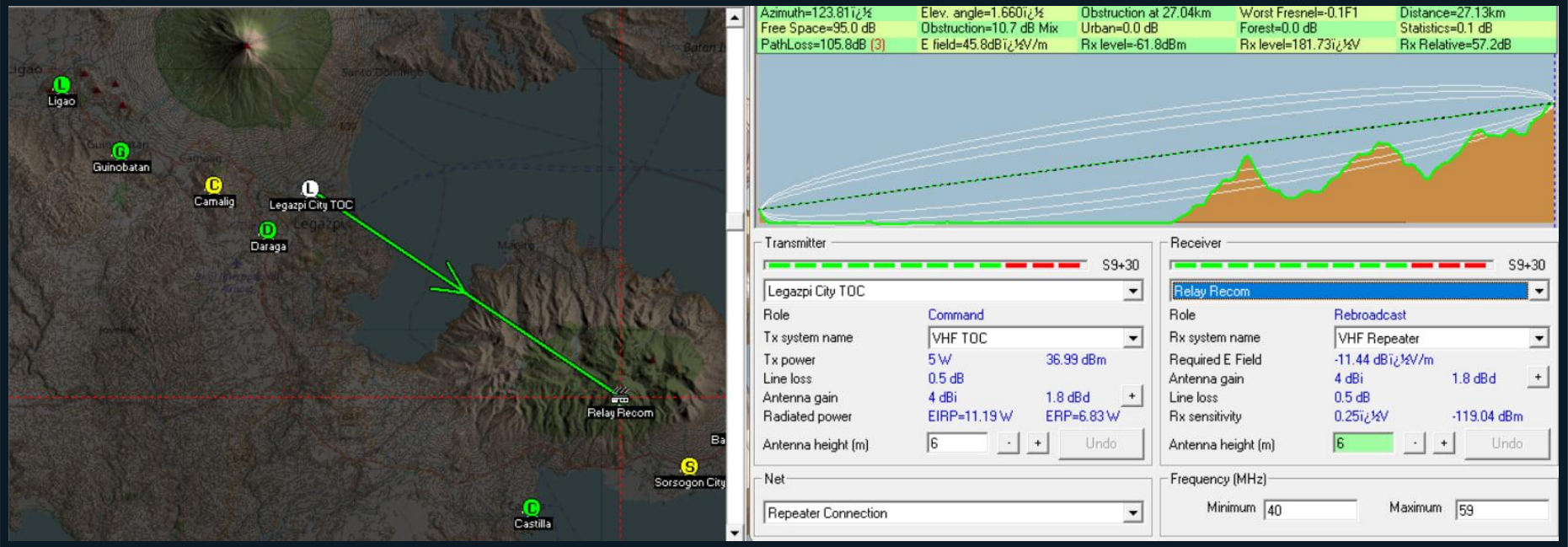
WORST FRESNEL
0.0 F1

LINK GO
Optical LOS - Fresnel zone partly clear

Radio Mobile Radio Link | Tx Ligao (Subordinate) > Rx Guinobatan (Rebroadcast) | Albay net

Repeater links: Bde HQ and Daraga reach Relay Alpha

LGP > RLY | DRG > RLY



DISTANCE
27.13 km

RX LEVEL
-61.8 dBm

FADE MARGIN
+57.2 dB

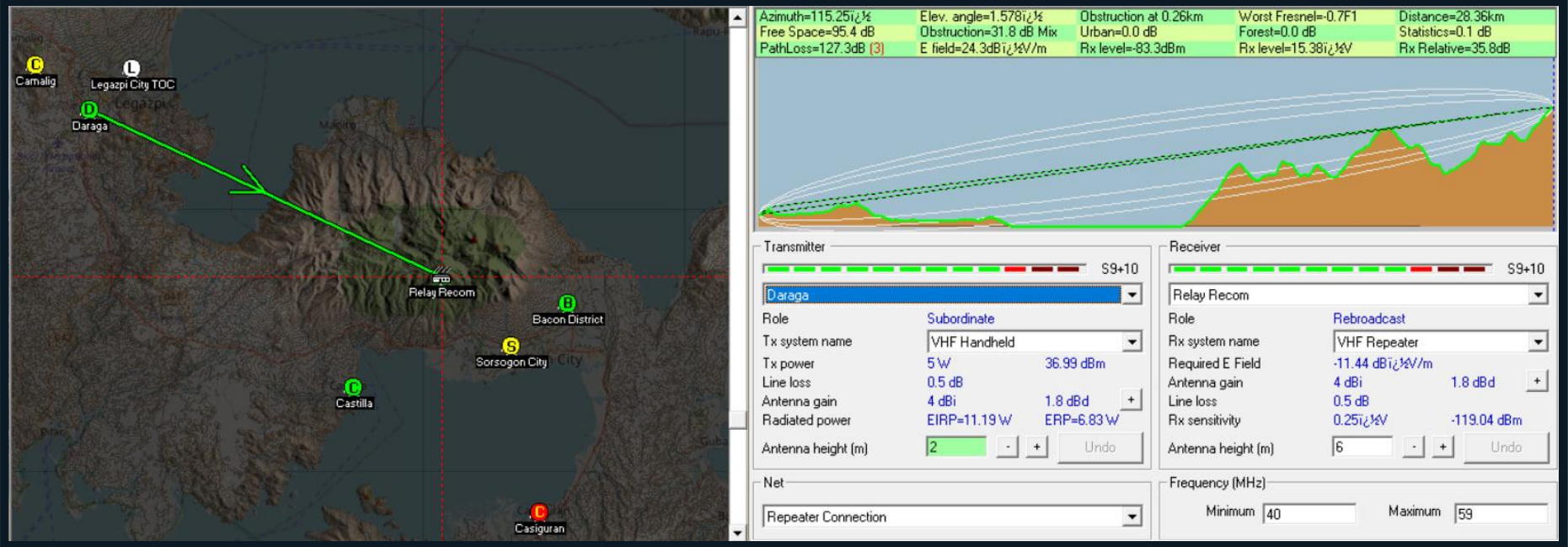
WORST FRESNEL
-0.1 F1

LINK GO
Terrain in path - closes by diffraction

Radio Mobile Radio Link | Tx Legazpi City TOC (Command) > Rx Relay Recom (Rebroadcast) | Repeater net

Repeater links: Bde HQ and Daraga reach Relay Alpha

LGP > RLY | DRG > RLY



DISTANCE
28.36 km

RX LEVEL
-83.3 dBm

FADE MARGIN
+35.8 dB

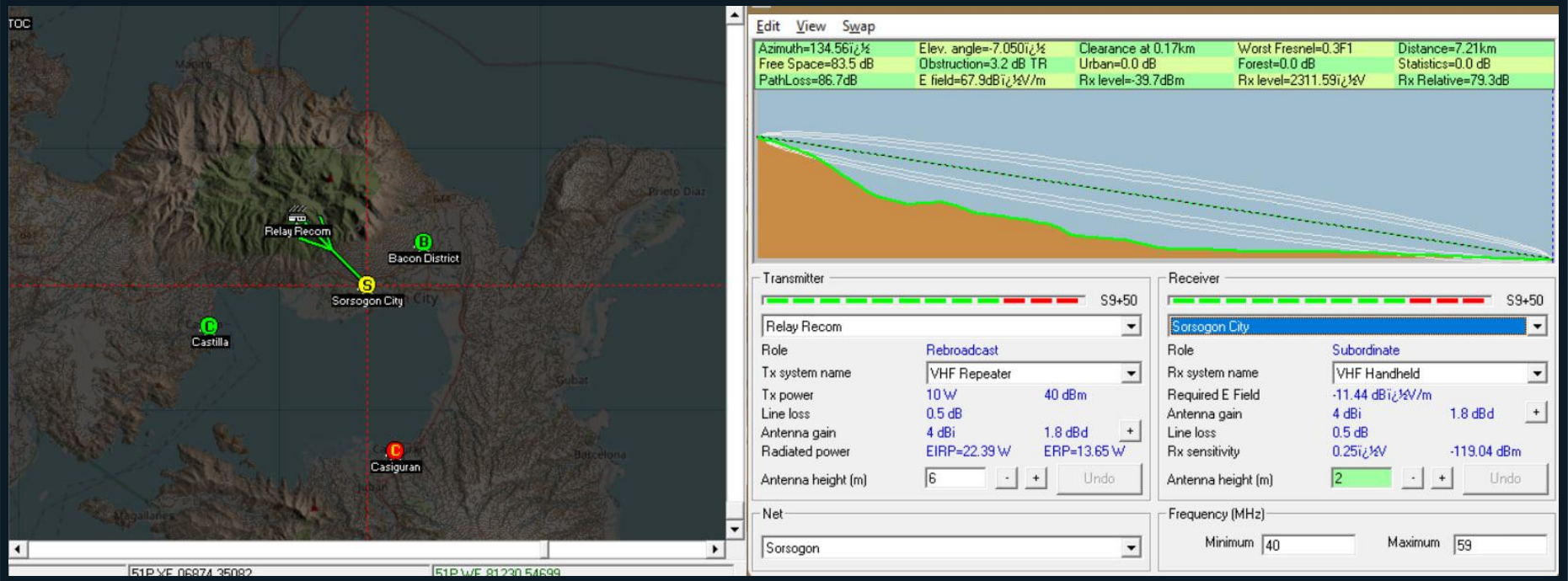
WORST FRESNEL
-0.7 F1

LINK GO
Terrain in path - closes by diffraction

Radio Mobile Radio Link | Tx Daraga (Subordinate) > Rx Relay Recom (Rebroadcast) | Repeater net

Sorsogon net: Relay Alpha reaches all four units

RLY > SOR | RLY > BAC | RLY > CST | RLY > CSG | SOR > BAC



DISTANCE
7.21 km

RX LEVEL
-39.7 dBm

FADE MARGIN
+79.3 dB

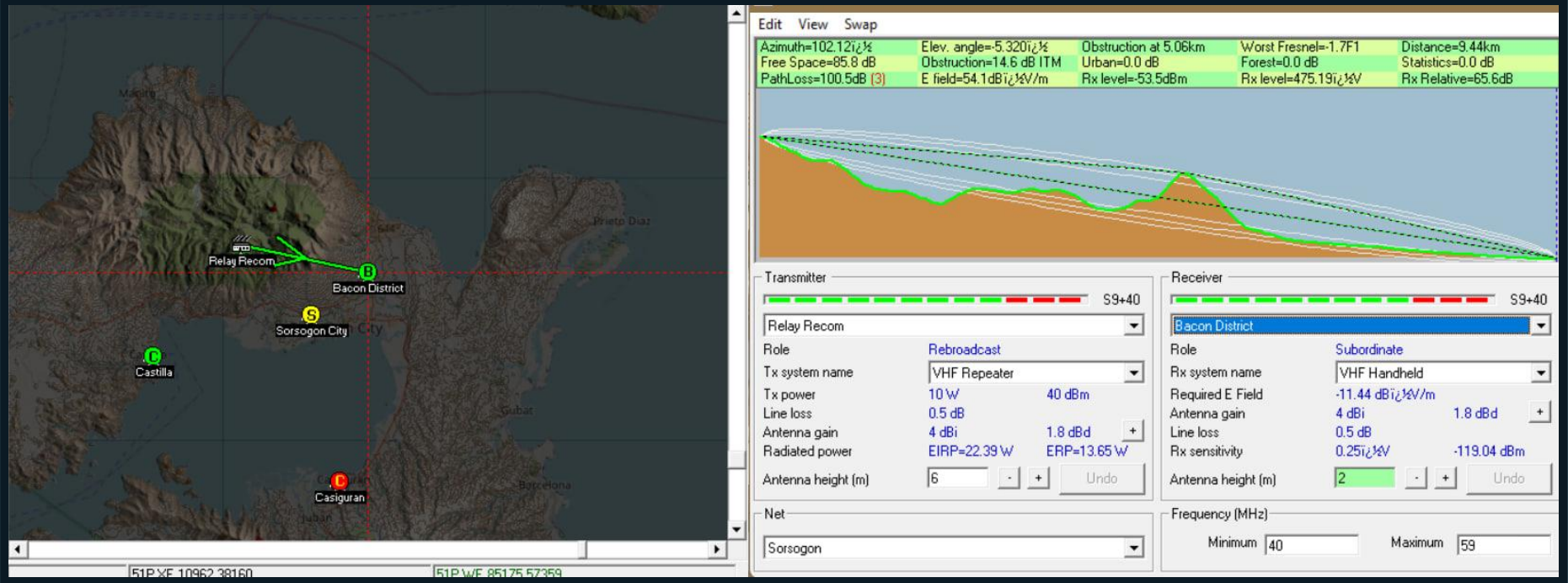
WORST FRESNEL
0.3 F1

LINK GO
Optical LOS - Fresnel zone partly clear

Radio Mobile Radio Link | Tx Relay Recom (Rebroadcast) > Rx Sorsogon City (Subordinate) | Sorsogon net

Sorsogon net: Relay Alpha reaches all four units

RLY > SOR | **RLY > BAC** | RLY > CST | RLY > CSG | SOR > BAC



DISTANCE
9.44 km

RX LEVEL
-53.5 dBm

FADE MARGIN
+65.6 dB

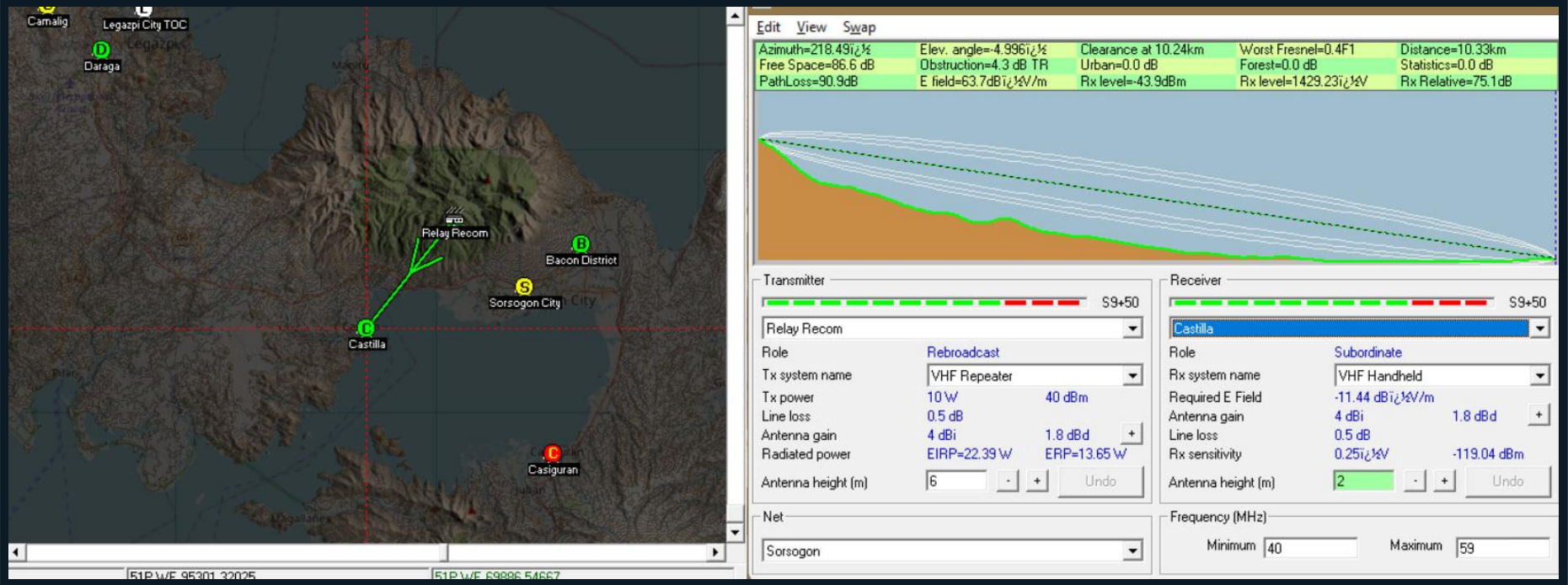
WORST FRESNEL
-1.7 F1

LINK GO
Terrain in path - closes by diffraction

Radio Mobile Radio Link | Tx Relay Recom (Rebroadcast) > Rx Bacon District (Subordinate) | Sorsogon net

Sorsogon net: Relay Alpha reaches all four units

RLY > SOR | RLY > BAC | **RLY > CST** | RLY > CSG | SOR > BAC



DISTANCE
10.33 km

RX LEVEL
-43.9 dBm

FADE MARGIN
+75.1 dB

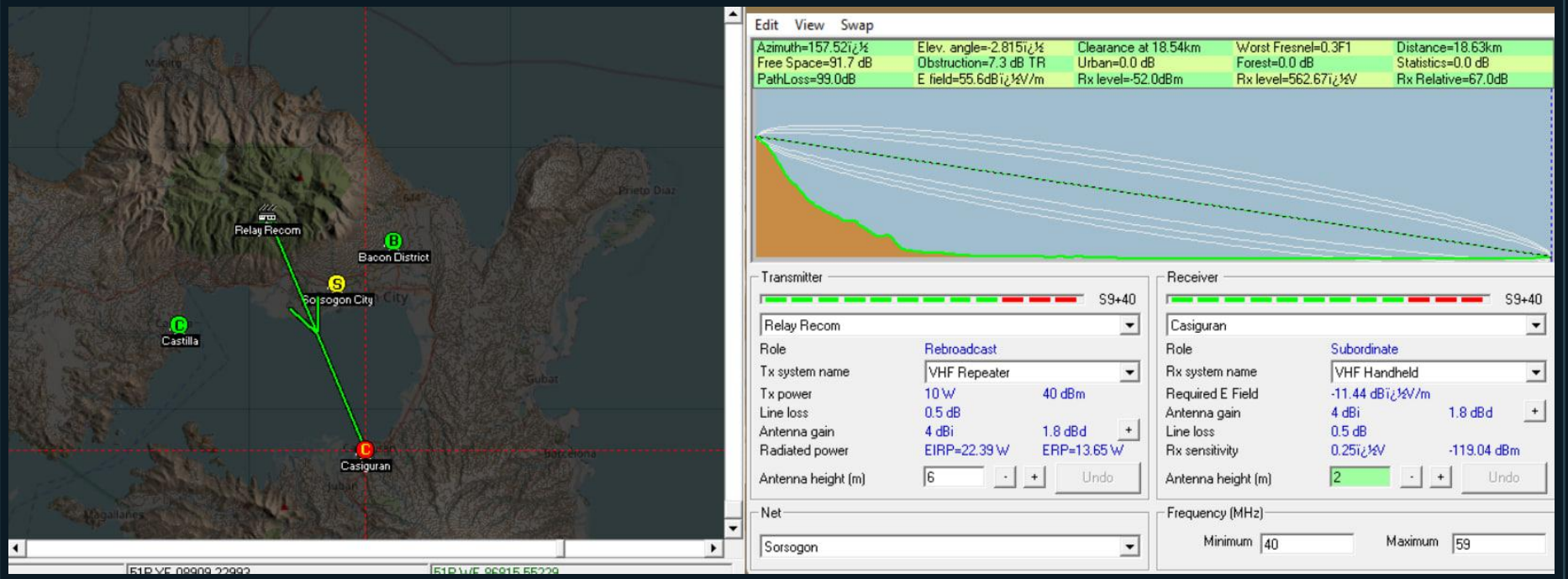
WORST FRESNEL
0.4 F1

LINK GO
Optical LOS - Fresnel zone partly clear

Radio Mobile Radio Link | Tx Relay Recom (Rebroadcast) > Rx Castilla (Subordinate) | Sorsogon net

Sorsogon net: Relay Alpha reaches all four units

RLY > SOR | RLY > BAC | RLY > CST | **RLY > CSG** | SOR > BAC



DISTANCE
18.63 km

RX LEVEL
-52.0 dBm

FADE MARGIN
+67.0 dB

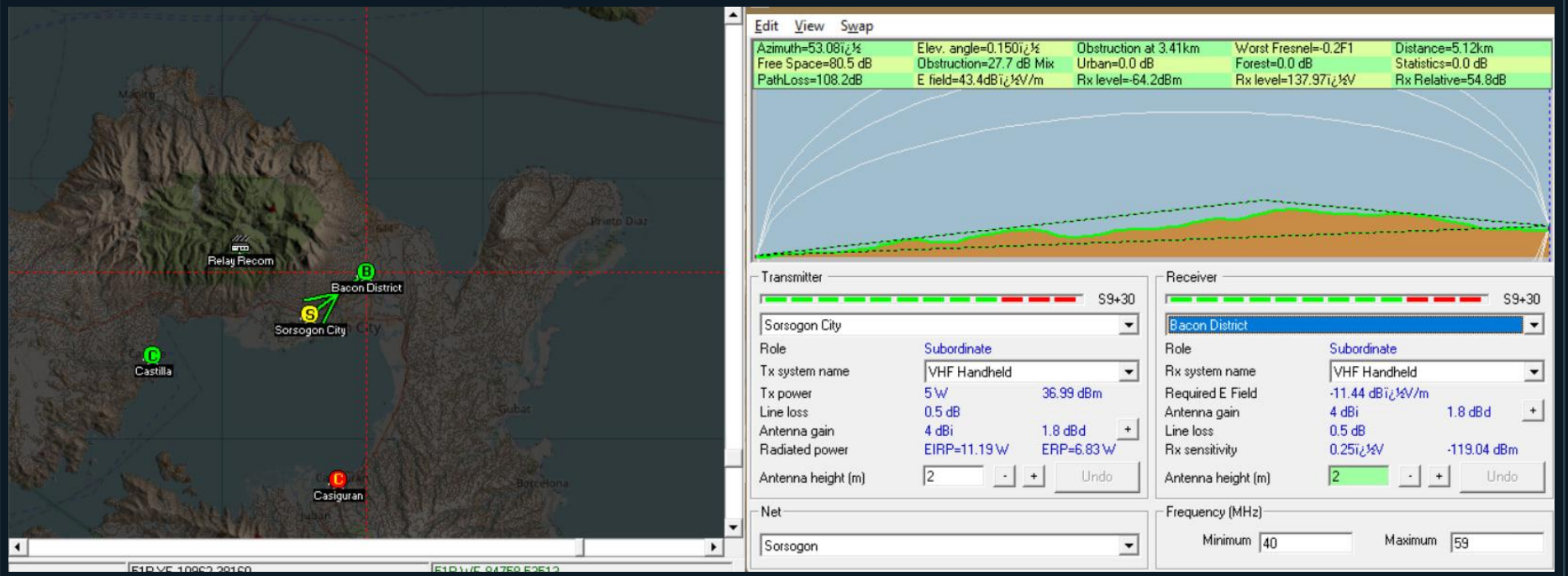
WORST FRESNEL
0.3 F1

LINK GO
Optical LOS - Fresnel zone partly clear

Radio Mobile Radio Link | Tx Relay Recom (Rebroadcast) > Rx Casiguran (Subordinate) | Sorsogon net

Sorsogon net: Relay Alpha reaches all four units

RLY > SOR | RLY > BAC | RLY > CST | RLY > CSG | **SOR > BAC**



DISTANCE
5.12 km

RX LEVEL
-64.2 dBm

FADE MARGIN
+54.8 dB

WORST FRESNEL
-0.2 F1

LINK GO
Terrain in path - closes by diffraction

Radio Mobile Radio Link | Tx Sorsogon City (Subordinate) > Rx Bacon District (Subordinate) | Sorsogon net

All 12 links close, 8 of them through terrain



Solid: optical LOS Dashed: terrain in path Colour: fade margin >= 60 / 40-60 / < 40 dB

LINK	KM	MARGIN	F1
LGP > DRG	4.3	+73.0	-0.1
LGP > CAM	7.1	+53.2	-0.8
DRG > CAM	5.2	+67.0	-0.3
CAM > GNB	7.2	+58.8	-0.8
LIG > GNB	6.5	+67.1	0.0
LGP > RLY	27.1	+57.2	-0.1
DRG > RLY	28.4	+35.8	-0.7
RLY > SOR	7.2	+79.3	0.3
RLY > BAC	9.4	+65.6	-1.7
RLY > CST	10.3	+75.1	0.4
RLY > CSG	18.6	+67.0	0.3
SOR > BAC	5.1	+54.8	-0.2

No path keeps the 60% Fresnel clearance a microwave link would need. At 40-59 MHz the signal bends over the ridges, so every link still holds 35.8 to 79.3 dB above the -119 dBm threshold.

SECTION 04 // STUDENT TASK 4

04

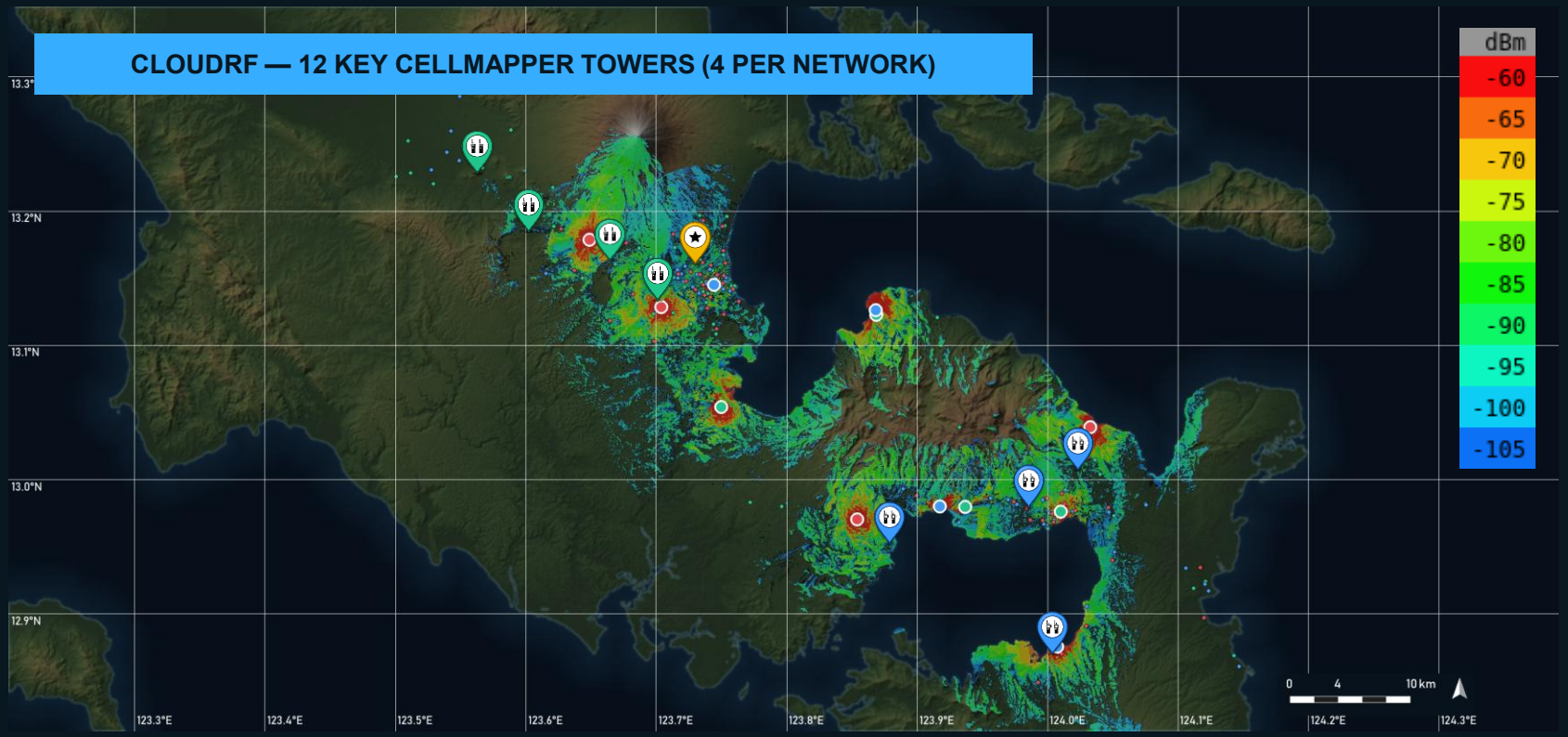
Radio & Cell Coverage

VHF from Radio Mobile; commercial cellular from CellMapper towers and CloudRF.



Photo: Jamesbourne1986, Wikimedia Commons, CC BY-SA 4.0

Cellular coverage — Smart, Globe and DITO



Large pins: towers run in CloudRF (Egli, 780 MHz, 20 W + 12 dBi, 30 m, tropical clutter; handset 1.5 m, −110 dBm). Small dots: other CellMapper towers (not run, 15-credit cap). Land only. Corridor covered by these 12: 51%.

CellMapper

Locate Smart (515-03), Globe (515-02) and DITO (515-66) towers in the AO

CloudRF

Multisite plot: tower height, band, EIRP; handset 1.5 m, −105 dBm

Google Earth Pro

Overlay KMZ with radio coverage; mark dead zones

Planning use

Cellular is the CONTINGENCY means only: towns and the coastal road are covered; the Pocdol interior and Mayon slopes are not. Use encrypted apps; never for classified traffic.

Inputs shared by the cellular exports

Input	Recorded value
Frequency	780 MHz
Transmit power	20 W
Transmit antenna gain	12 dBi
Transmit antenna height	30 m AGL
Receive antenna height	1.5 m
Receiver threshold	−110 dBm

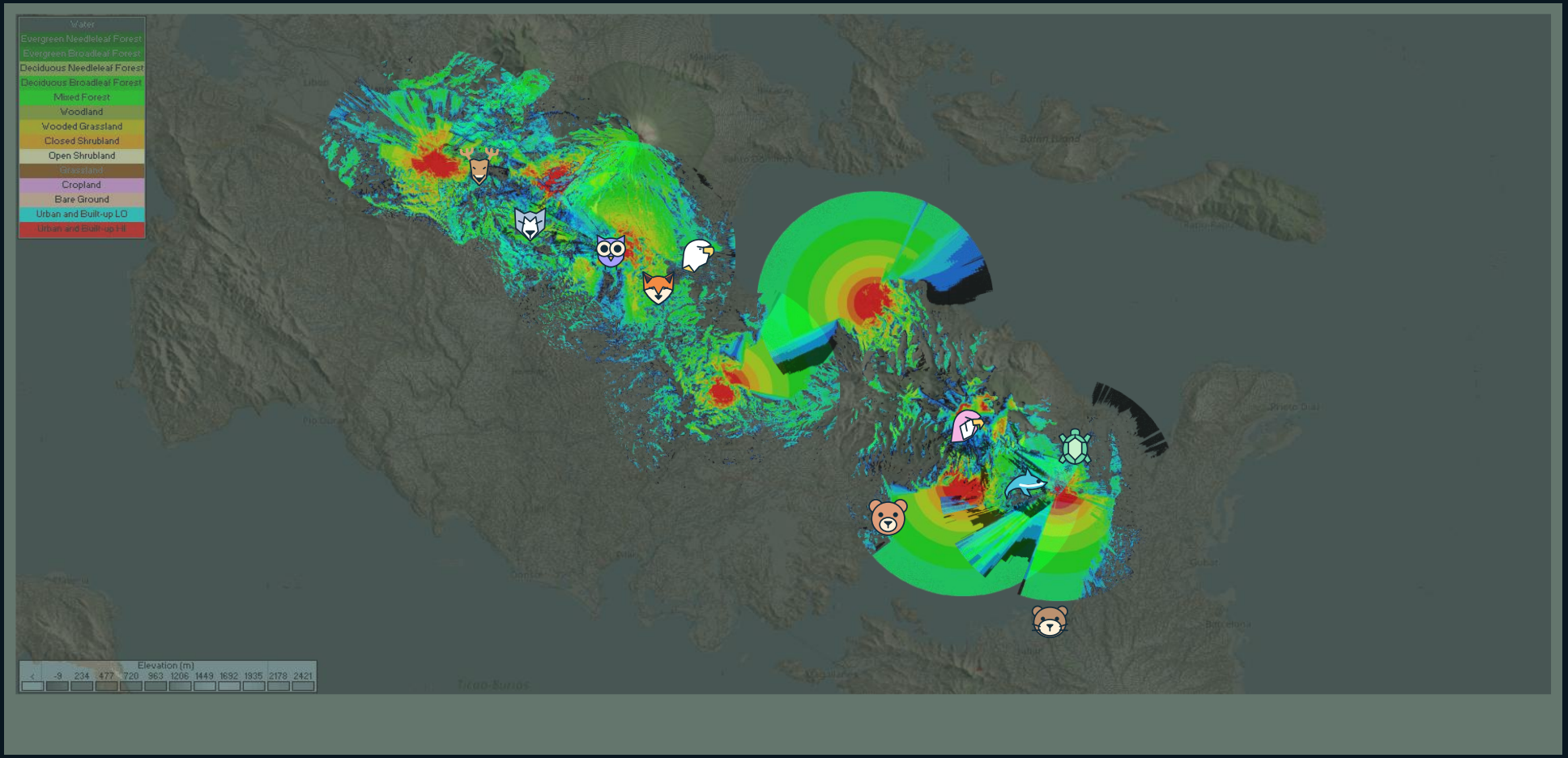
22

supplied models

Matching inputs
in all 22 requests

Model inputs describe these exports, not the operators’ verified equipment or service.

SMART



8

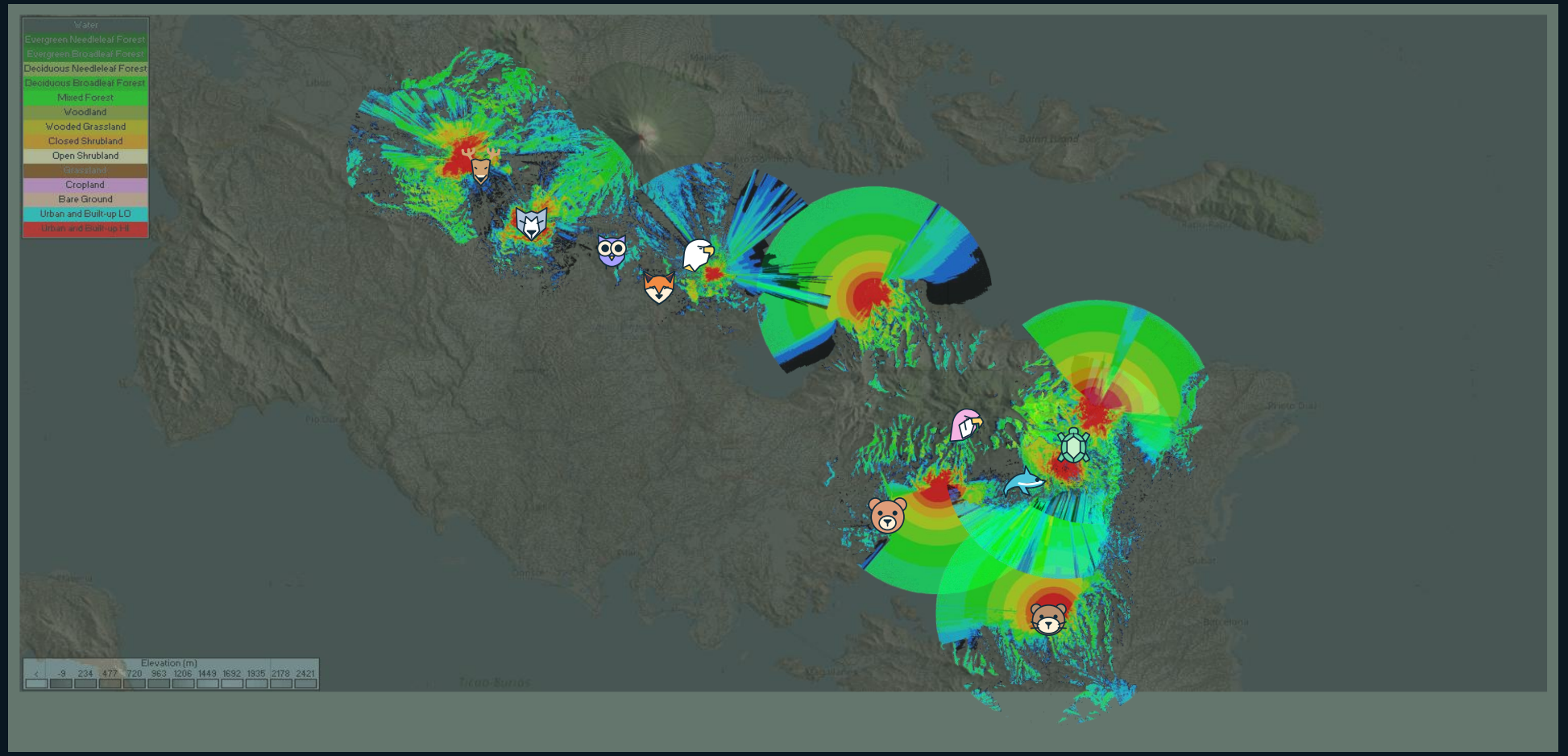
saved models

SMART 01
SMART 02
SMART 03
SMART 04
SMART 05
SMART 06
SMART 07
SMART 08

dBm
-60
-65
-70
-75
-80
-85
-90
-95
-100
-105

SMART uses the rabbit marker. Raster colours retain the original signal legend.

GLOBE



8

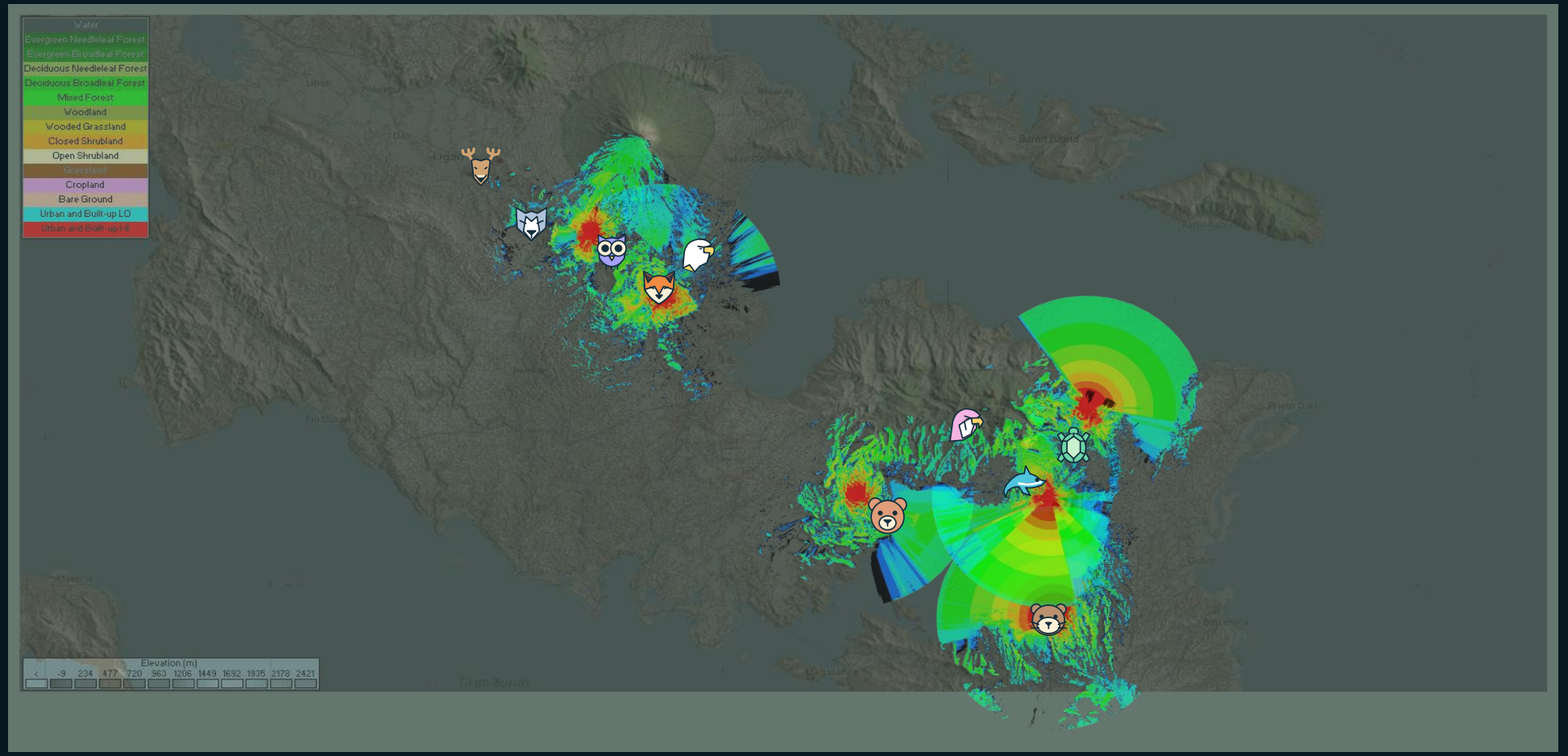
saved models

GLOBE 01
GLOBE 02
GLOBE 03
GLOBE 04
GLOBE 05
GLOBE 06
GLOBE 07
GLOBE 08

dBm
-60
-65
-70
-75
-80
-85
-90
-95
-100
-105

GLOBE uses the penguin marker. Raster colours retain the original signal legend.

DITO



6

saved models

DITO 01
DITO 02
DITO 03
DITO 04
DITO 05
DITO 06

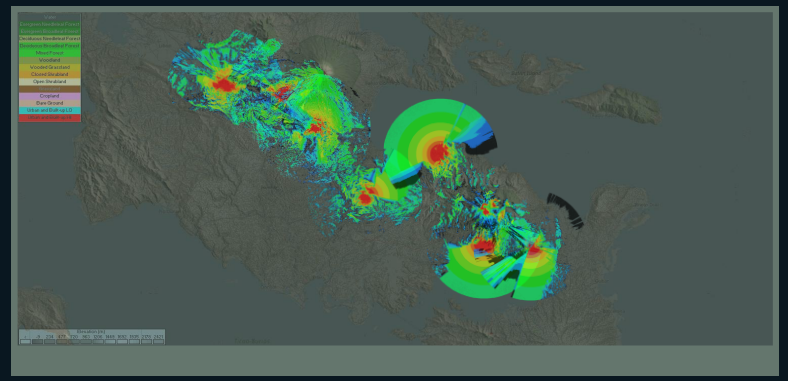
dBm
-60
-65
-70
-75
-80
-85
-90
-95
-100
-105

DITO uses the elephant marker. Raster colours retain the original signal legend.

COMMON MAP EXTENT

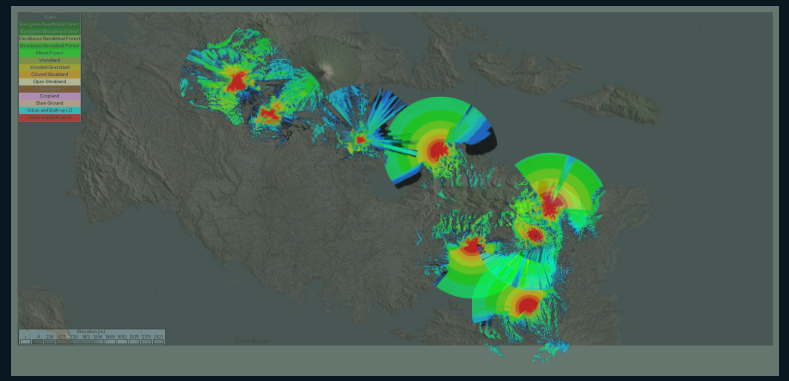
Cellular coverage comparison

SMART



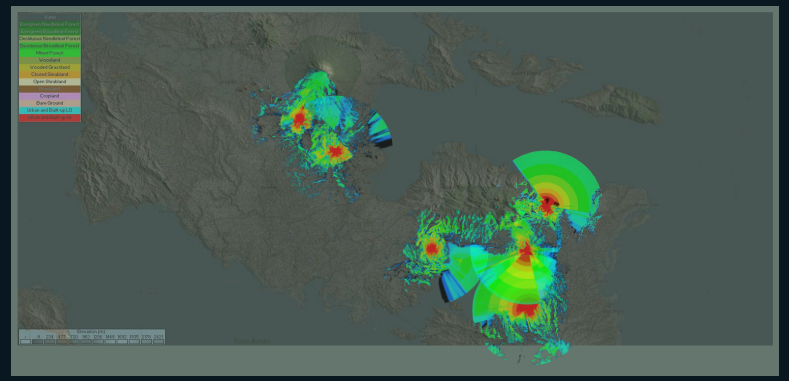
8 supplied models

GLOBE



8 supplied models

DITO



6 supplied models

The same map extent supports comparison. It does not establish complete service coverage.



SECTION 05 // STUDENT TASK 5

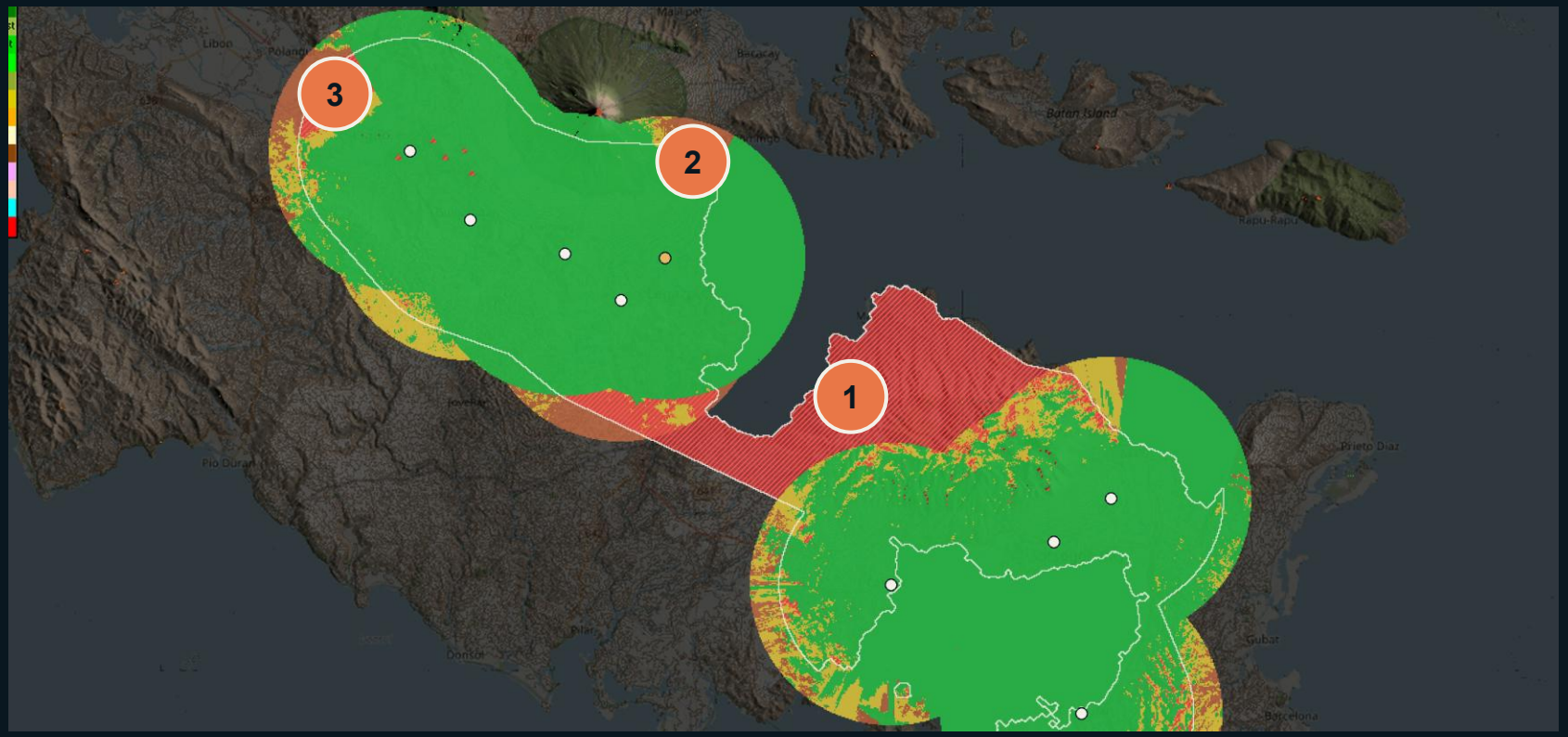
05

Communication Dead Spots

Where the operational corridor has no usable VHF signal.

Photo: Rauxilijari, Wikimedia Commons, CC BY-SA 4.0

Three dead spots — one splits the brigade



Red hatch = corridor land with no plotted signal or only the weakest Radio Mobile classes (pink and red). White outline = land within 8 km of the 12 planned links. Recomputed from the Radio Mobile exports, no relay.

DS-1 179 km²

Daraga–Castilla seam (Manito slopes)
Outside every station's 10 km plot; the Pocdol foothills separate the clusters.
51P WQ 92505 45181

DS-2 6 km²

Sto. Domingo / Mayon SE
Lower volcano slopes shadow Legazpi's NE sector.
51P WQ 81130 61773

DS-3 14 km²

Oas / Ligao NW
Ridge NW of Ligao; edge of the 10 km plot.
51P WQ 55640 66505

SECTION 06 // STUDENT TASK 6

06

Relay Station

Covering the Daraga to Castilla gap for handhelds and mobiles.

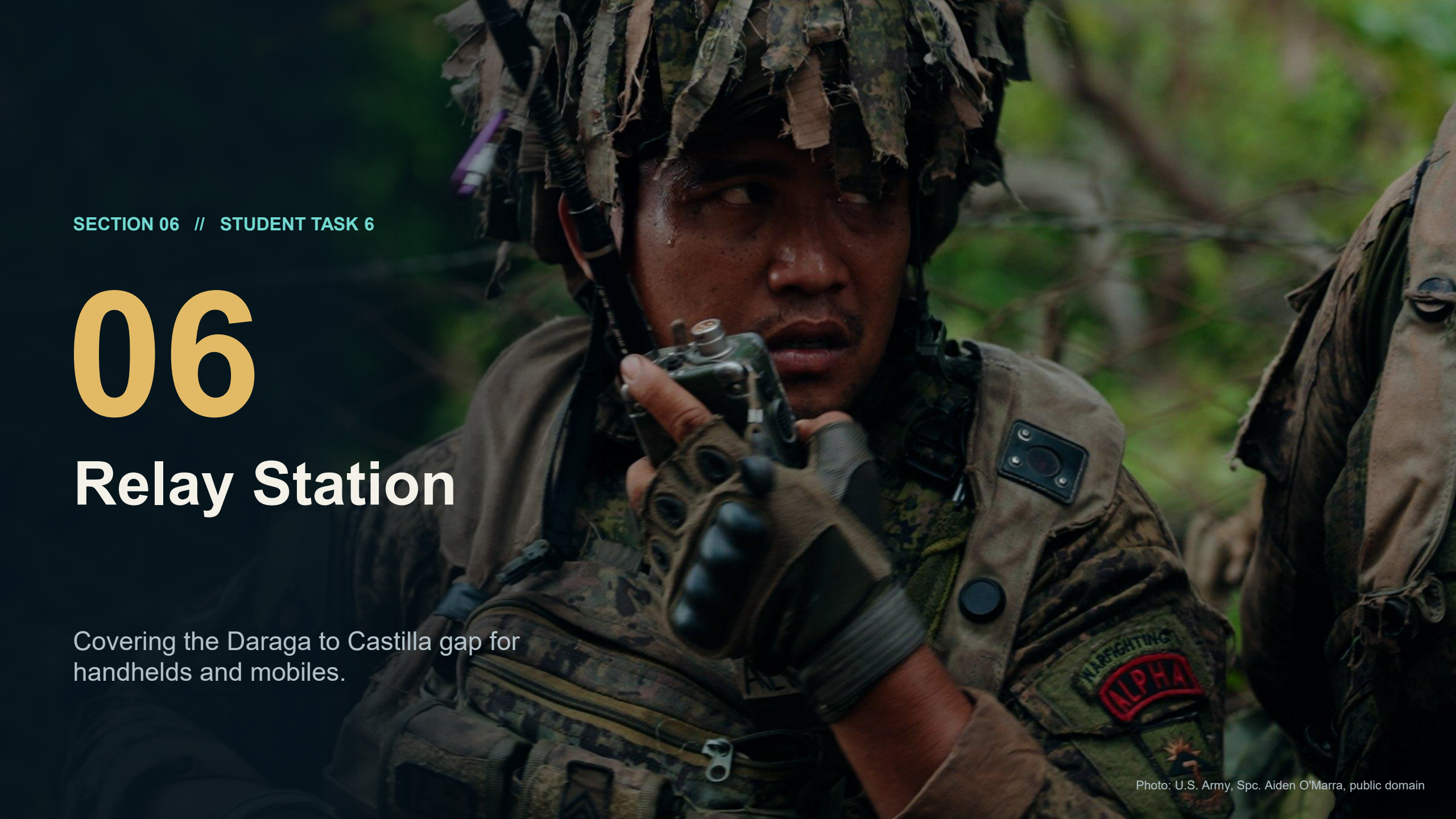
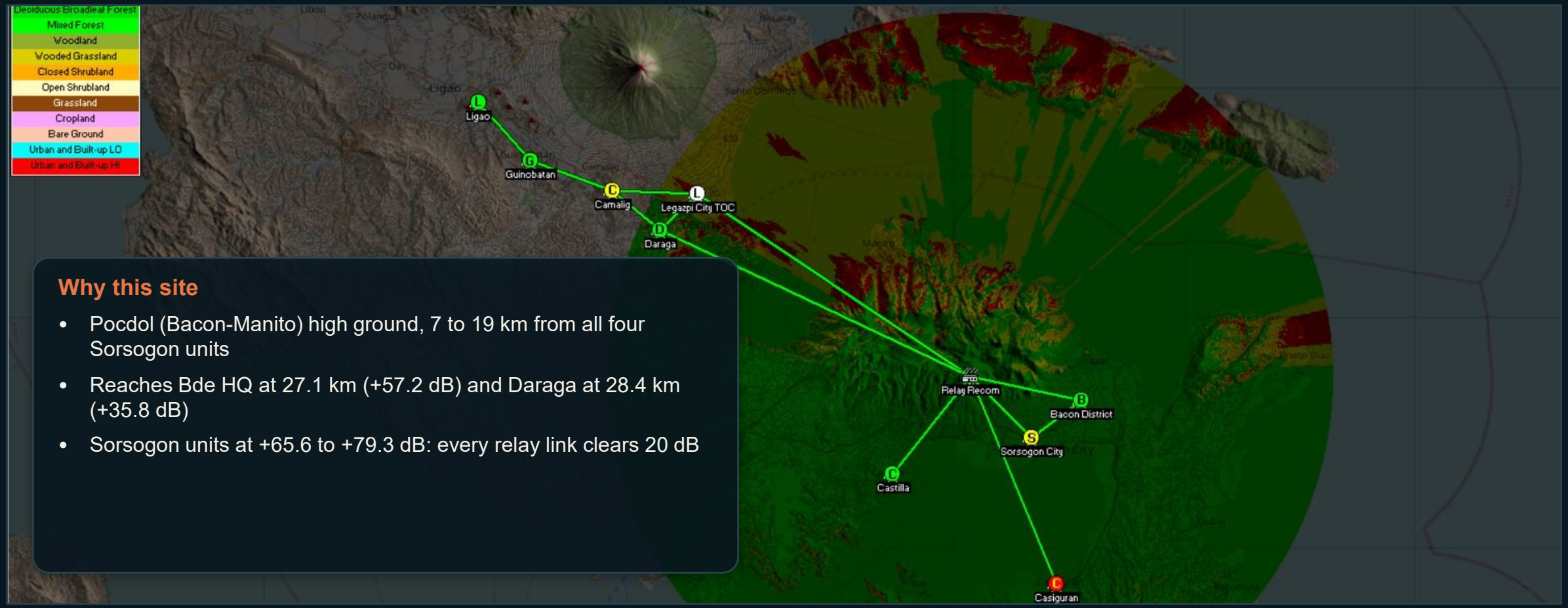


Photo: U.S. Army, Spc. Aiden O'Marra, public domain

RELAY ALPHA — bridging the Daraga–Castilla gap

GRID (MGRS)	ELEVATION	LINKS ≥ 20 dB
51P XQ 01711 40097	931 m + 6 m mast	6 of 6



Why this site

- Pocdol (Bacon-Manito) high ground, 7 to 19 km from all four Sorsogon units
- Reaches Bde HQ at 27.1 km (+57.2 dB) and Daraga at 28.4 km (+35.8 dB)
- Sorsogon units at +65.6 to +79.3 dB: every relay link clears 20 dB

06 // RELAY SITE

Relay Alpha site: Pocdol high ground, beside an existing compound



GRID

51P XQ 01711 40097

GROUND

931 m above sea level

ANTENNA

6 m above ground

ACCESS

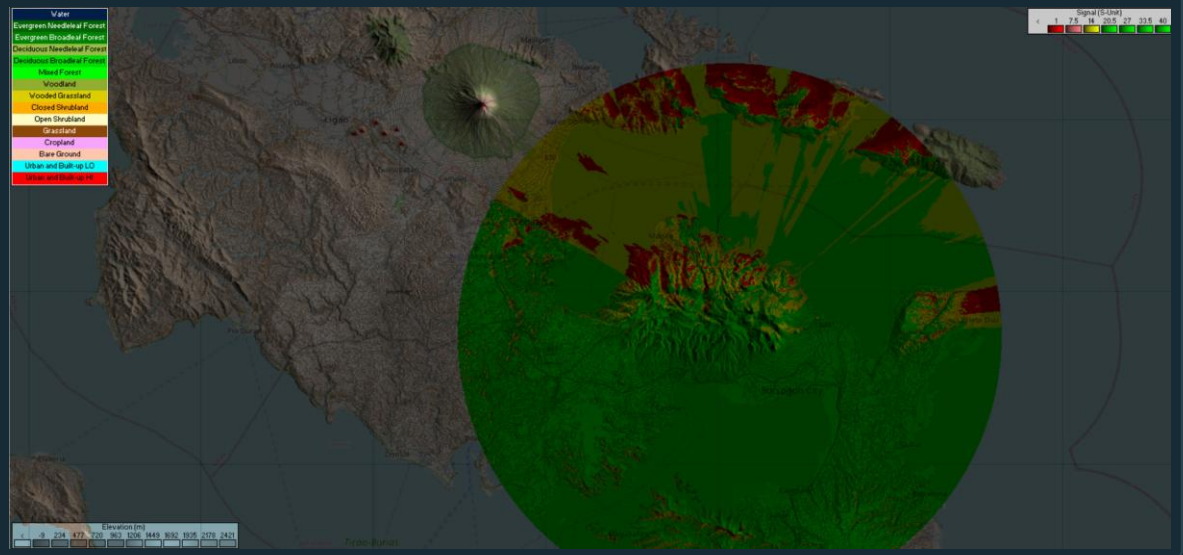
Road and cleared compound beside the site

Google Earth view, imagery 6 Apr 2026 (Airbus, Maxar Technologies). Lines are the planned links.

Relay Alpha coverage in Radio Mobile

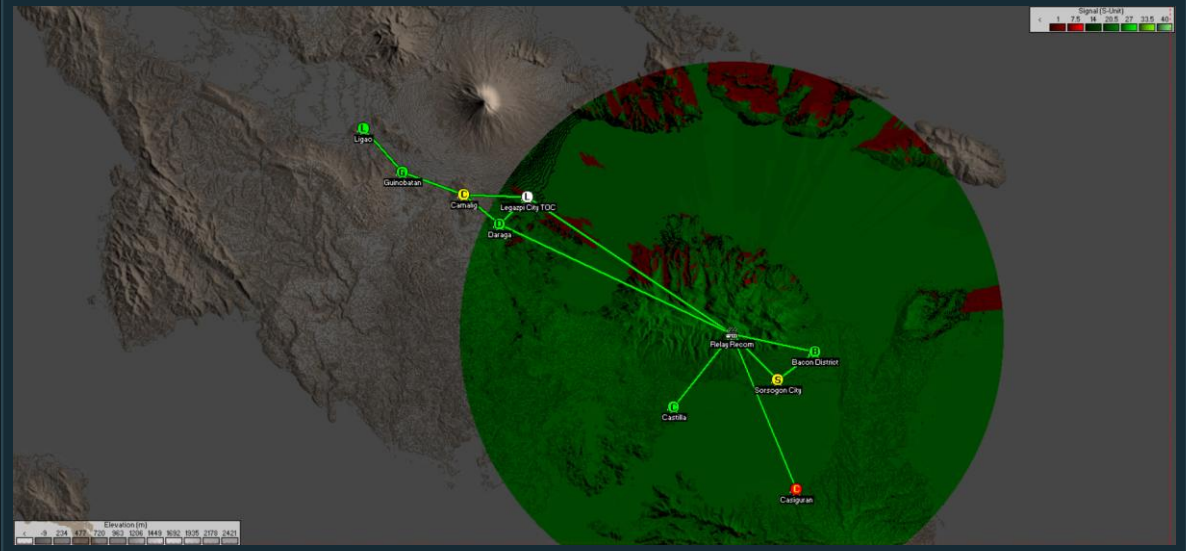
Radio Mobile - Relay Recom coverage, 10 W at 6 m

SCREENSHOT



Radio Mobile - Relay Recom with its links to all units

SCREENSHOT



51P XQ 01711 40097

Repeater grid

6 m

Antenna height

10 W

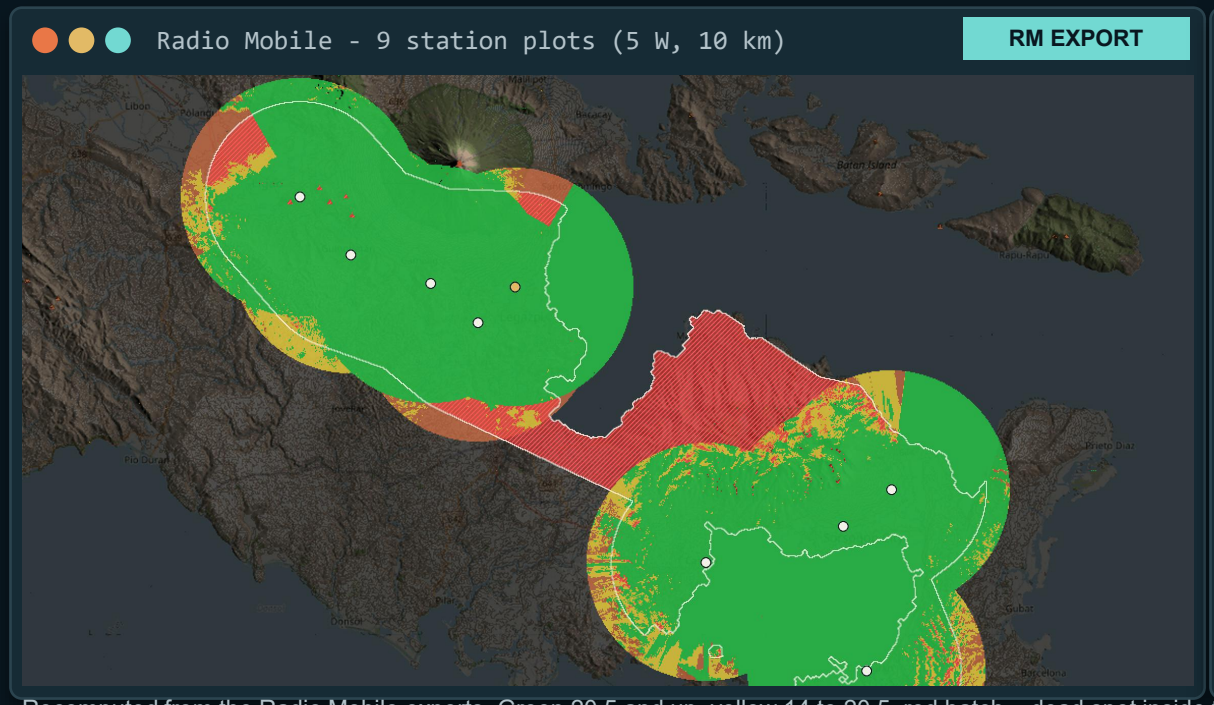
Tx power (in / out)

6 / 6

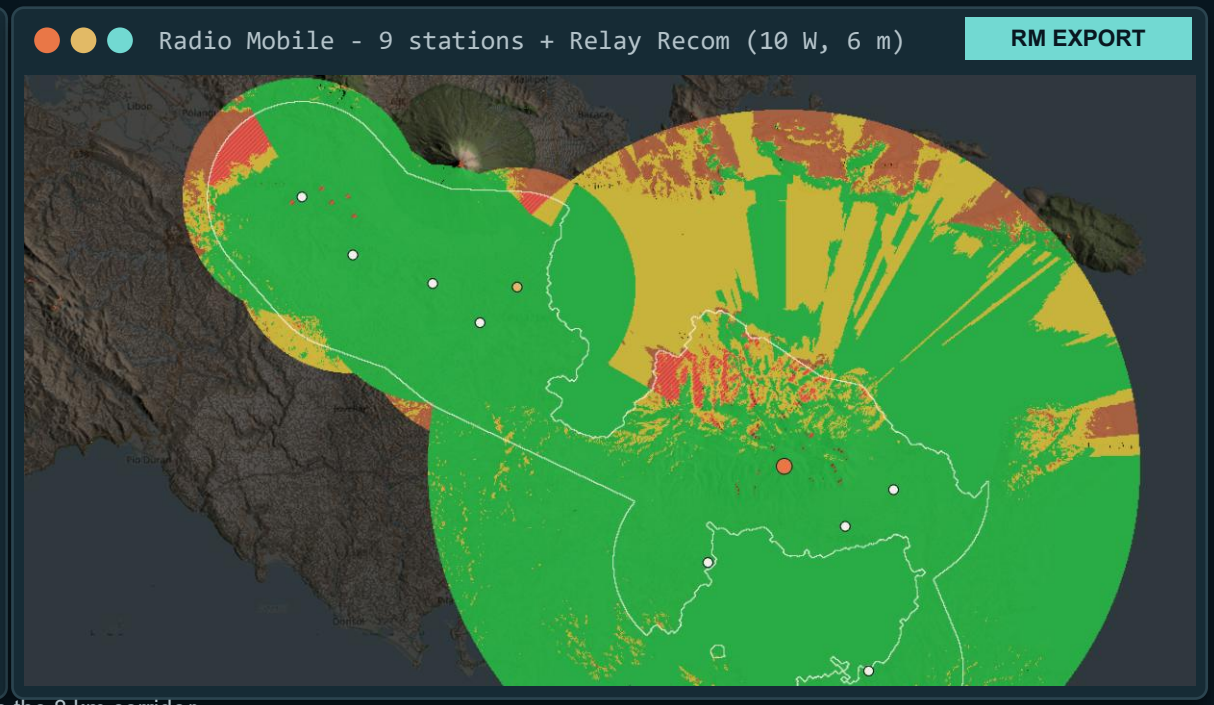
Relay links >= 20 dB

Relay Alpha shrinks DS-1 and joins both basins

BEFORE — OVERALL (5 W)



AFTER - WITH RELAY ALPHA (10 W)



Recomputed from the Radio Mobile exports. Green 20.5 and up, yellow 14 to 20.5, red hatch = dead spot inside the 8 km corridor.

81% → 96%

Corridor with usable VHF

179 → 15 km²

DS-1 dead-spot area

2 → 1

Radio basins (one brigade net)

12 / 12

Links >= 20 dB fade margin

SECTION 07 // STUDENT TASKS 7-8

07

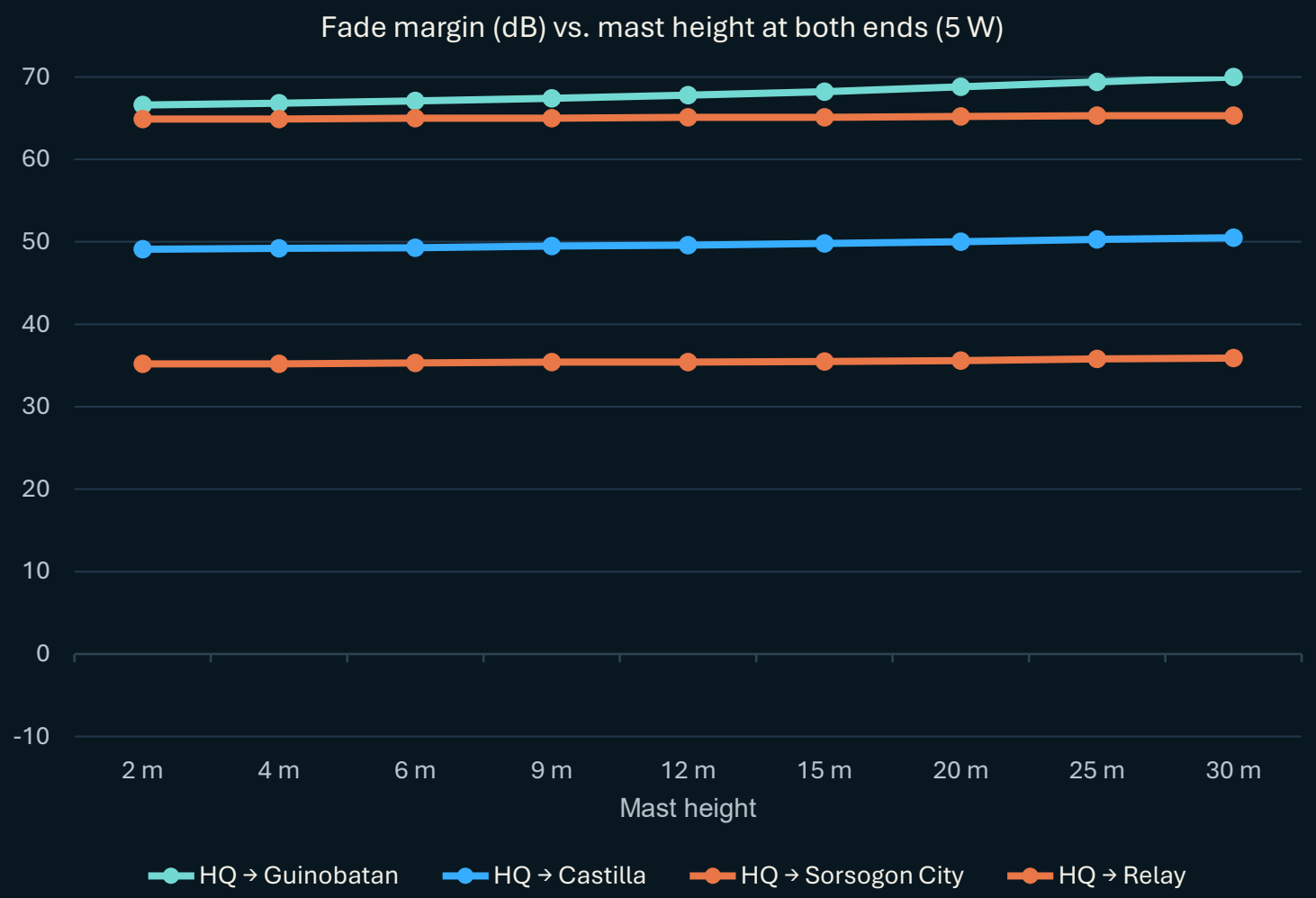
Antenna Height & Tx Power

Smallest mast and lowest power that still give 20 dB of fade margin.



Photo: CPO Kimberly Martinez, U.S. Navy (public domain)

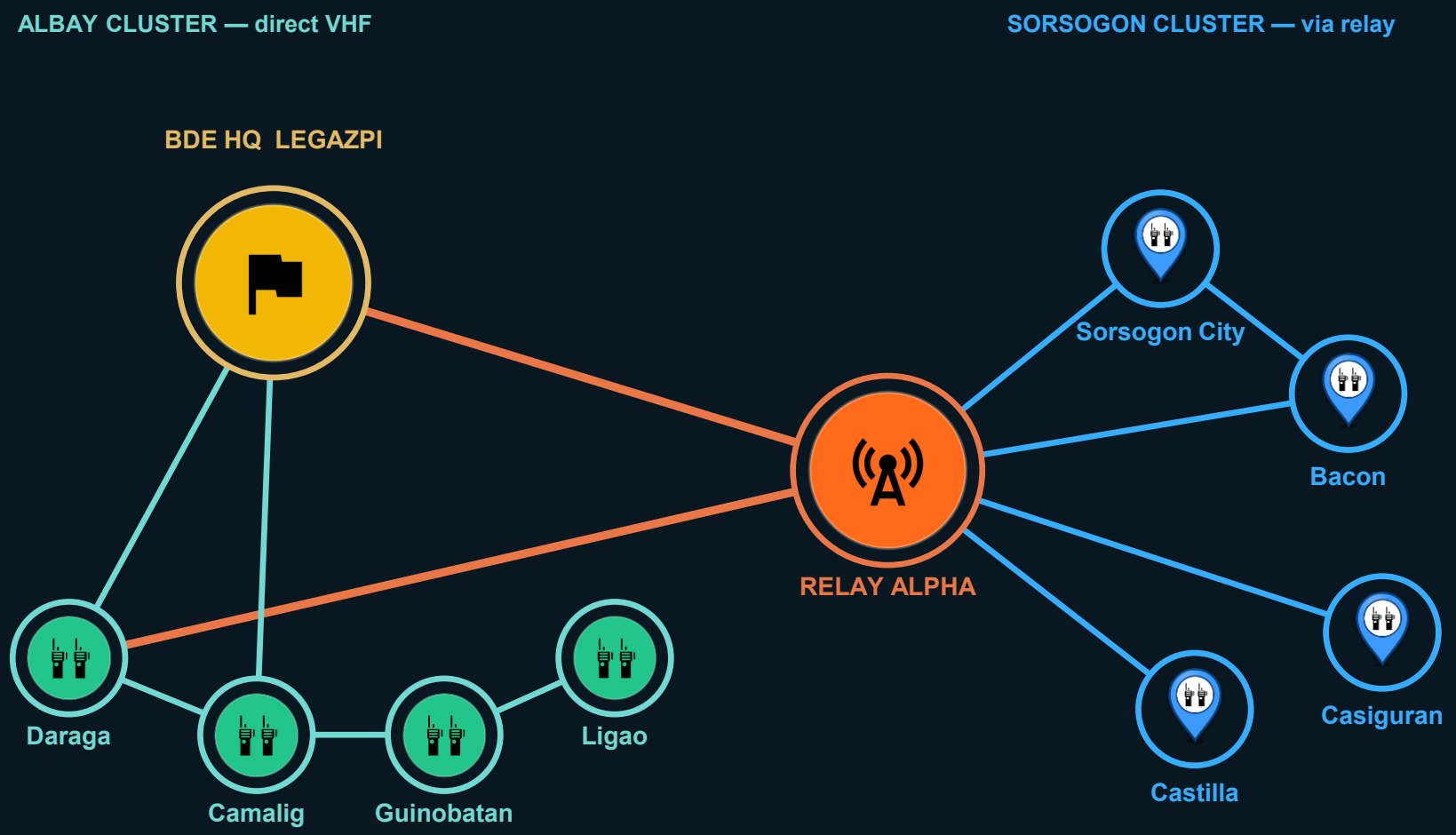
Mast height barely matters at 40–59 MHz



Station	Mast	Why
Bde HQ (Legazpi)	**6 m**	As modelled in Radio Mobile; clears 20 dB on every link
Relay Alpha	**6 m**	931 m ground on Pocdol already gives the height
Unit stations (8)	**2 m**	Handheld height, as modelled
Dismounted	**1.8 m**	Handheld; relay within reach of most areas

Insight
Raising HQ–Sorsogon City masts from 2 m to 30 m only moves the margin from 35.2 to 35.9 dB. At 6 m it already closes; use height for handheld reach, not base links.

One brigade net via Relay Alpha + PACE



P — PRIMARY
VHF FM net via Relay Alpha (retrans)

A — ALTERNATE
HF NVIS voice/data, Bde ↔ units

C — CONTINGENCY
Cellular (Smart / Globe / DITO), encrypted apps

E — EMERGENCY
SATCOM terminal / runner

Camalig and Guinobatan rebroadcast for Ligao. Sorsogon City and Bacon also work each other direct.

Recommendations to the Brigade Commander

Emplace Relay Alpha

51P XQ 01711 40097 on Pocdol high ground: 10 W retrans with a security element. Bde HQ +57.2 dB, Daraga +35.8 dB, Sorsogon units +54.8 dB or better.

Set antenna heights

Bde HQ and Relay Alpha on 6 m masts; unit handhelds at 2 m, as modelled in Radio Mobile. Height does not cross the Pocdol range; the relay does.

Set transmit power

5 W base stations (every base link ≥ 23 dB at 5 W); repeater 10 W; Harris handhelds 5 W, Sorsogon side through the repeater.

Manage dead spots

Relay Alpha cuts DS-1 from 179 to 15 km² (Manito slopes remain). DS-2 Sto. Domingo and DS-3 Oas: move to high ground or use HF NVIS; report before entering.

Use cellular as contingency

CloudRF on 12 key CellMapper towers: towns and coastal road covered, Pocdol interior not.

Validate on the ground

Recon relay access (BacMan geothermal roads), conduct radio checks on all 12 links, update plots.

Questions?

Reliable comms. Uninterrupted C2.

1LT ERIC JAN B INTOY (SC) PA • Signal Officer, 1BCT • SOAC 52-2026



Sorsogon Bay panorama. Photo: Irvin Parco Sto. Tomas, Wikimedia Commons, CC BY-SA 4.0